

AIM:

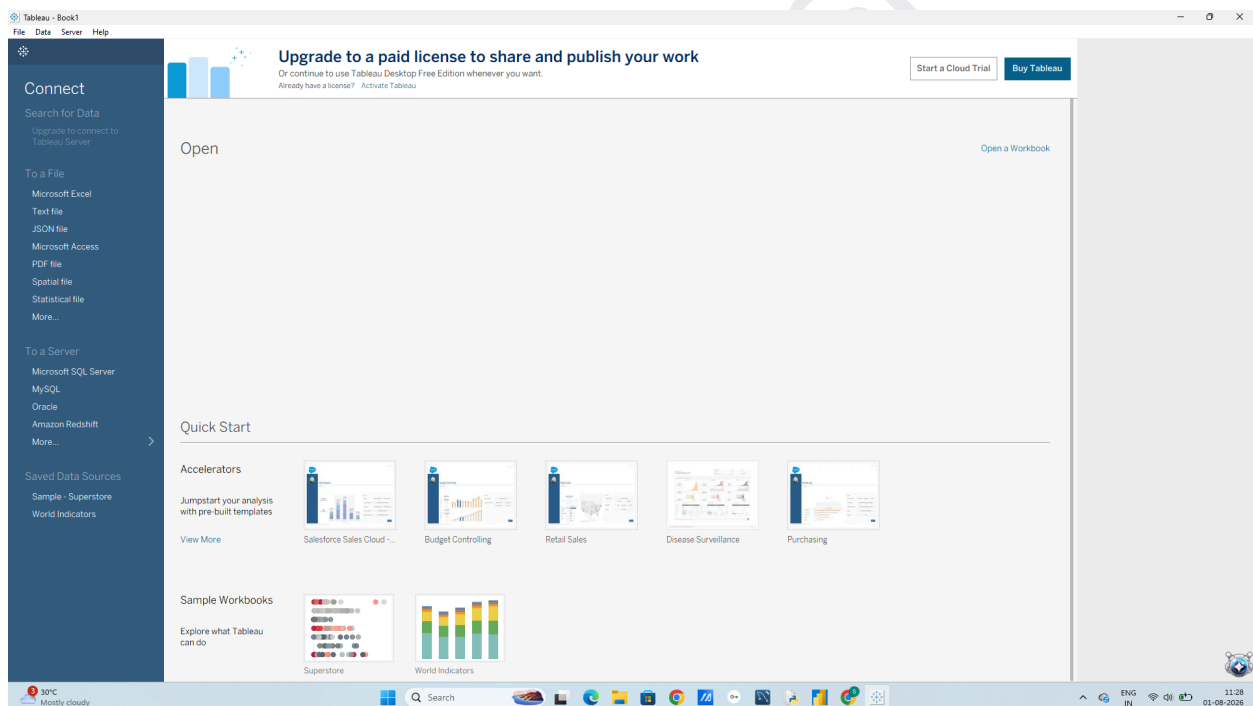
Lesson 1: Getting Started with Tableau and Building Basic Visualizations

- Connect to a CSV dataset and explore the Tableau interface
- Identify and differentiate Measures and Dimensions
- Convert fields between discrete and continuous and observe changes
- Create and customize a basic bar chart
- Build a line chart for time-series analysis
- Create a geographic map visualization
- Combine multiple charts into a simple interactive dashboard

Dataset: Superstore.

kaggle link: <https://www.kaggle.com/code/rajatraj0502/superstore-dataset>

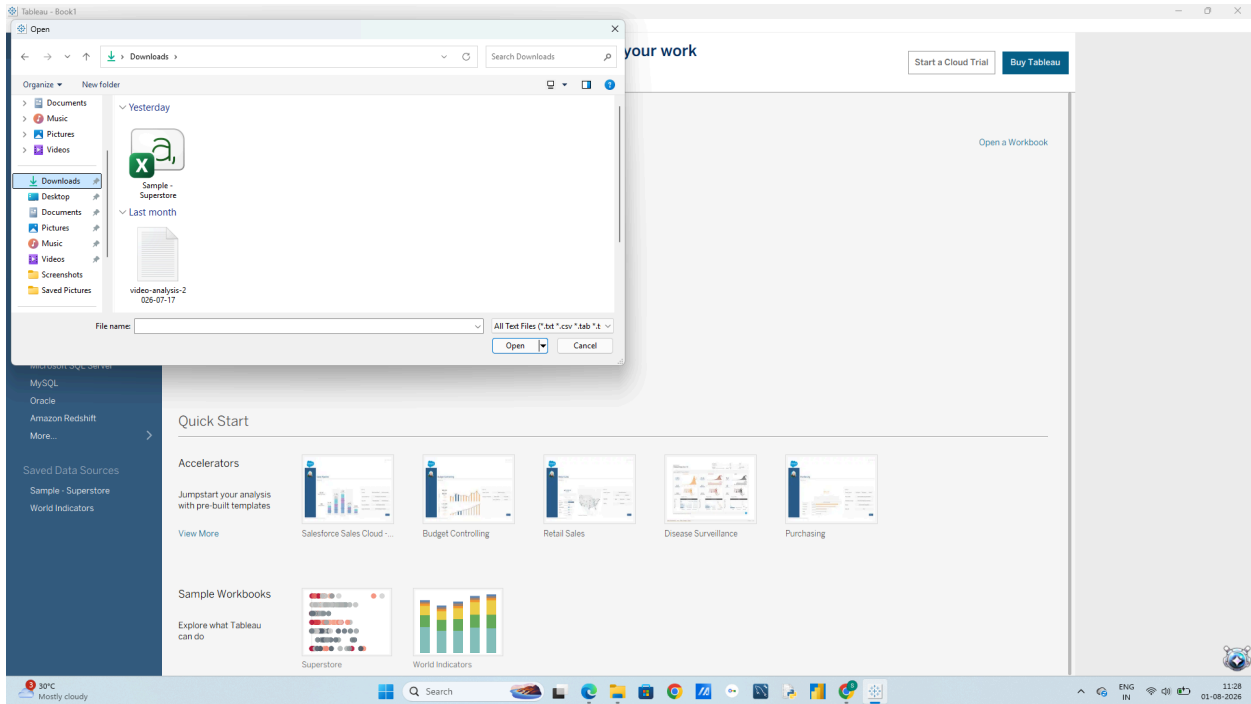
Part A: Connect to Dataset & Explore the Interface 1. Launch Tableau Desktop on your machine



NAME: ADARSH SHUKLA

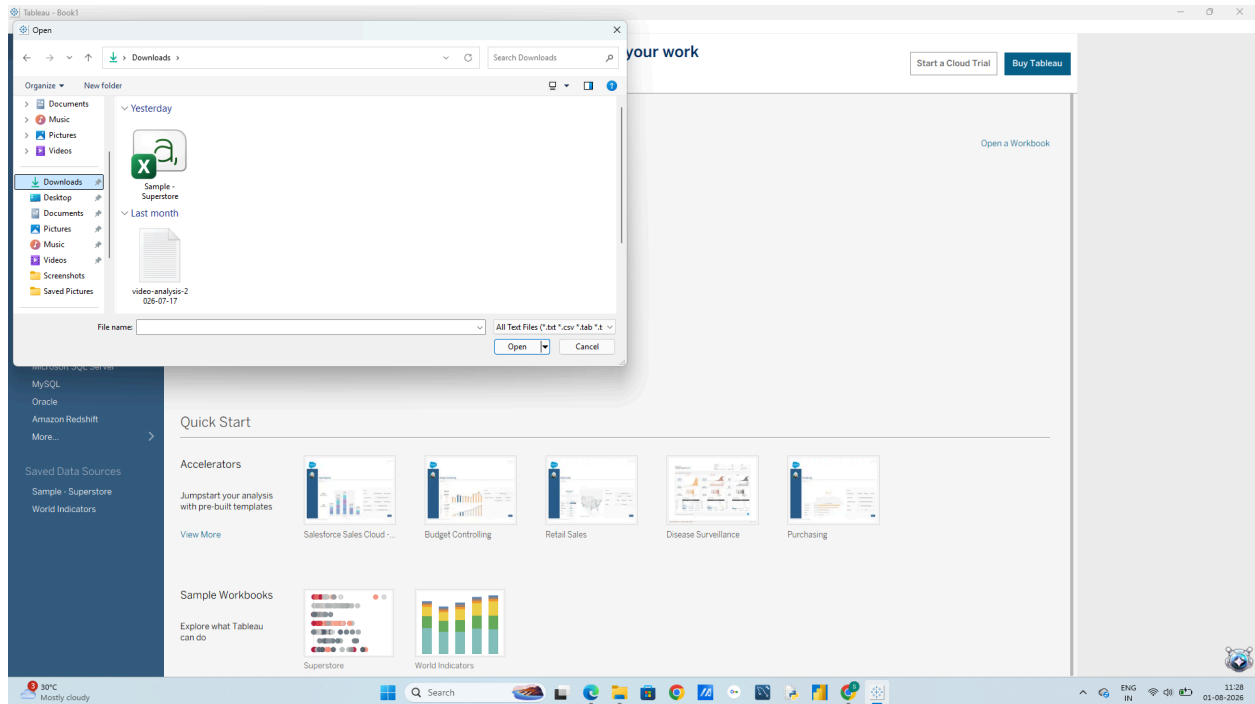
ROLL NO : T117

2. On the left-hand blue navigation pane under Connect, click on Text File (Tableau treats CSVs as text files) which is marked in a red circle.



3. Select your CSV file from your Mac Finder and click Open.

Practical 1 :1: **Getting Started with Tableau and Building Basic Visualizations**



4. You will be taken to the Data Source page. Here, preview your data rows at the bottom to ensure they look correct, then click Sheet 1 at the bottom left to enter your workspace marked in a red circle inside the snapshot. Tableau will bypass its faulty drivers, create a native workspace called Clipboard, and instantly open up Sheet

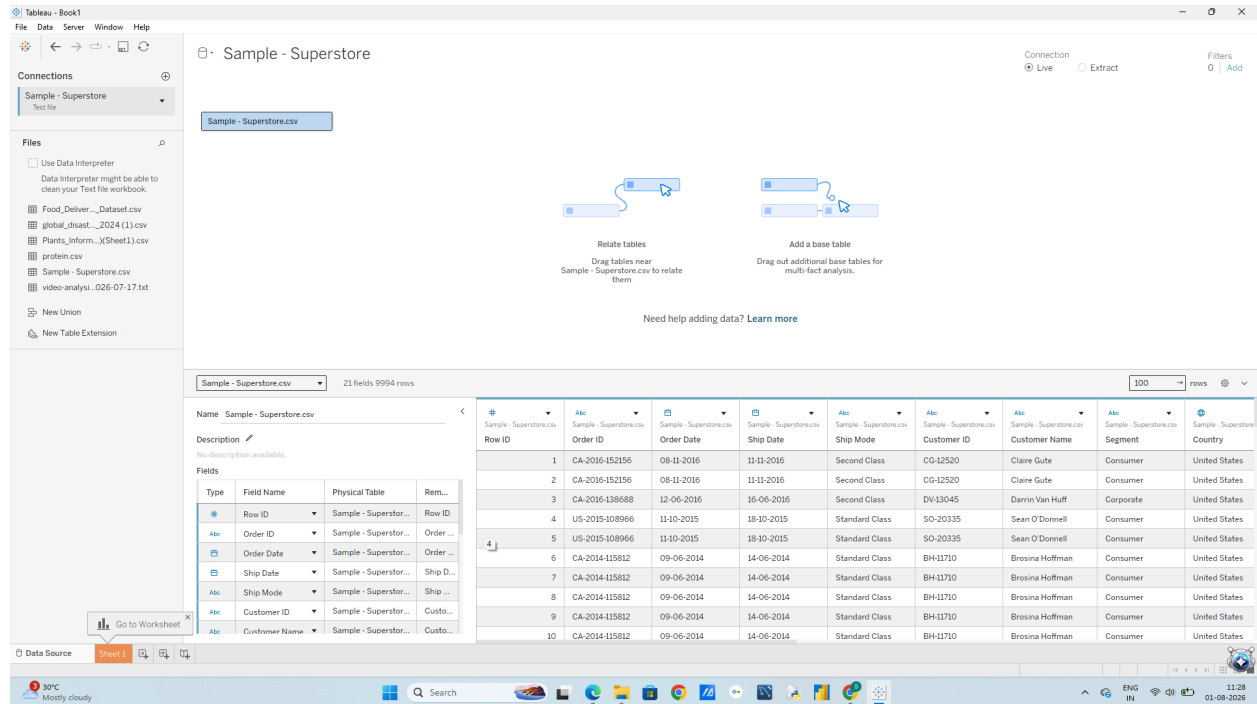
NAME: ADARSH SHUKLA

ROLL NO : T117

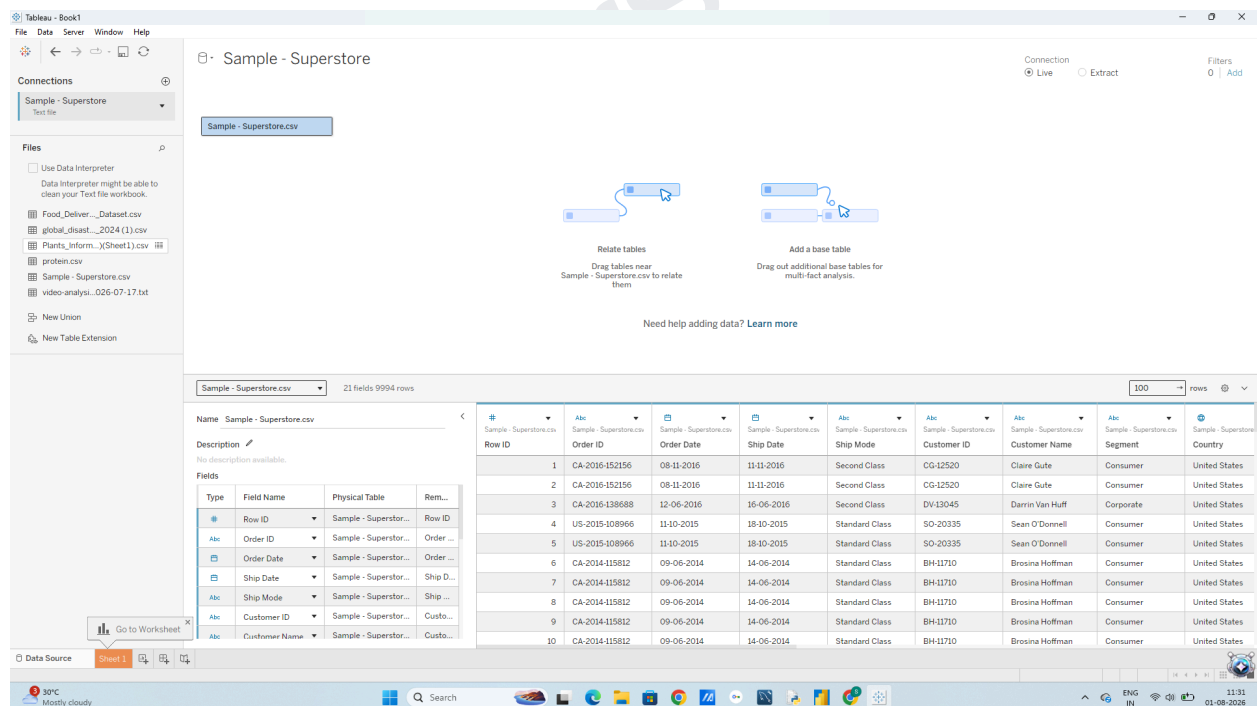
SHETH L.U.J. AND SIR M.V. COLLEGE OF ARTS, SCIENCE AND COMMERCE

Subject: :Subject: Data Visualization with Power BI and Tableau

Practical 1 :1: Getting Started with Tableau and Building Basic Visualizations



Navigating the Interface The Data Pane (Far Left): This holds all your variables from the dataset



Shelves (Top/Center): The Columns and Rows shelves act as your X and Y coordinates.

NAME: ADARSH SHUKLA

ROLL NO : T117

SHETH L.U.J. AND SIR M.V. COLLEGE OF ARTS, SCIENCE AND COMMERCE

Subject: :Subject: Data Visualization with Power BI and Tableau

Practical 1 :1: Getting Started with Tableau and Building Basic Visualizations

The screenshot displays the Tableau Desktop interface for a 'Sample - Superstore' data source. The interface is divided into several sections:

- Connections:** A sidebar on the left showing the 'Sample - Superstore' connection.
- Files:** A sidebar on the left showing a list of files, including 'Food_Deliver...Dataset.csv', 'global_disast...2024 (1).csv', 'Plants_inform...y(Sheet1).csv', 'proten.csv', 'Sample - Superstore.csv', and 'video-analysi...026-07-17.txt'.
- Central Workspace:** A large area in the center showing a data preview table. The table has 10 columns: Row ID, Order ID, Order Date, Ship Date, Ship Mode, Customer ID, Customer Name, Segment, and Country. The data is filtered to show 10 rows.
- Bottom Status Bar:** A bar at the bottom showing the current data source, 'Sample - Superstore.csv', and the number of rows, '21 fields 9994 rows'.

The data preview table shows the following data:

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country
1	CA-2016-152156	08-11-2016	11-11-2016	Second Class	CG-12520	Claire Gule	Consumer	United States
2	CA-2016-152156	08-11-2016	11-11-2016	Second Class	CG-12520	Claire Gule	Consumer	United States
3	CA-2016-138688	12-06-2016	16-06-2016	Second Class	DV-13045	Darrin Van Huff	Corporate	United States
4	US-2015-108966	11-10-2015	18-10-2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States
5	US-2015-108966	11-10-2015	18-10-2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States
6	CA-2014-115812	09-06-2014	14-06-2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States
7	CA-2014-115812	09-06-2014	14-06-2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States
8	CA-2014-115812	09-06-2014	14-06-2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States
9	CA-2014-115812	09-06-2014	14-06-2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States
10	CA-2014-115812	09-06-2014	14-06-2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States

The Marks Card (Left of center): This controls your visual design properties like color, text labels, and sizes.

NAME: ADARSH SHUKLA
ROLL NO : T117

SHETH L.U.J. AND SIR M.V. COLLEGE OF ARTS, SCIENCE AND COMMERCE

Subject: :Subject: Data Visualization with Power BI and Tableau

Practical 1 :1: Getting Started with Tableau and Building Basic Visualizations

Tableau - Book1

File Data Server Window Help

Connections

Sample - Superstore
Text file

Files

Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

Food_Deliver_...Dataset.csv
global_dias...2024 (1).csv
Plants_inform...y(Sheet1).csv
proten.csv
Sample - Superstore.csv
video-analysi...026-07-17.txt

New Union
New Table Extension

Sample - Superstore.csv

Connection
Live Extract
Filters
Add

Relate tables
Drag tables near Sample - Superstore.csv to relate them

Add a base table
Drag out additional base tables for multi-fact analysis.

Need help adding data? [Learn more](#)

Sample - Superstore.csv 21 fields 9994 rows 100 rows

Name Sample - Superstore.csv
Description
No description available.

Fields

Type	Field Name	Physical Table	Rem...
Row ID	Row ID	Sample - Superstor...	Row ID
Order ID	Order ID	Sample - Superstor...	Order ...
Order Date	Order Date	Sample - Superstor...	Order ...
Ship Date	Ship Date	Sample - Superstor...	Ship D...
Ship Mode	Ship Mode	Sample - Superstor...	Ship ...
Customer ID	Customer ID	Sample - Superstor...	Custo...
Customer Name	Customer Name	Sample - Superstor...	Custo...

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country
1	CA-2016-152156	08-11-2016	11-11-2016	Second Class	CG-12520	Claire Gule	Consumer	United States
2	CA-2016-152156	08-11-2016	11-11-2016	Second Class	CG-12520	Claire Gule	Consumer	United States
3	CA-2016-138688	12-06-2016	16-06-2016	Second Class	DV-13045	Darrin Van Huff	Corporate	United States
4	US-2015-108966	11-10-2015	18-10-2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States
5	US-2015-108966	11-10-2015	18-10-2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States
6	CA-2014-115812	09-06-2014	14-06-2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States
7	CA-2014-115812	09-06-2014	14-06-2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States
8	CA-2014-115812	09-06-2014	14-06-2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States
9	CA-2014-115812	09-06-2014	14-06-2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States
10	CA-2014-115812	09-06-2014	14-06-2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States

Go to Worksheet

Data Source: Sample - Superstore.csv

11:31 01-08-2028

Part B: Identify and Differentiate Measures and Dimensions

Look closely at your Data Pane on the left. Tableau reads the metadata of the Superstore CSV and splits your items automatically:

NAME: ADARSH SHUKLA

ROLL NO : T117

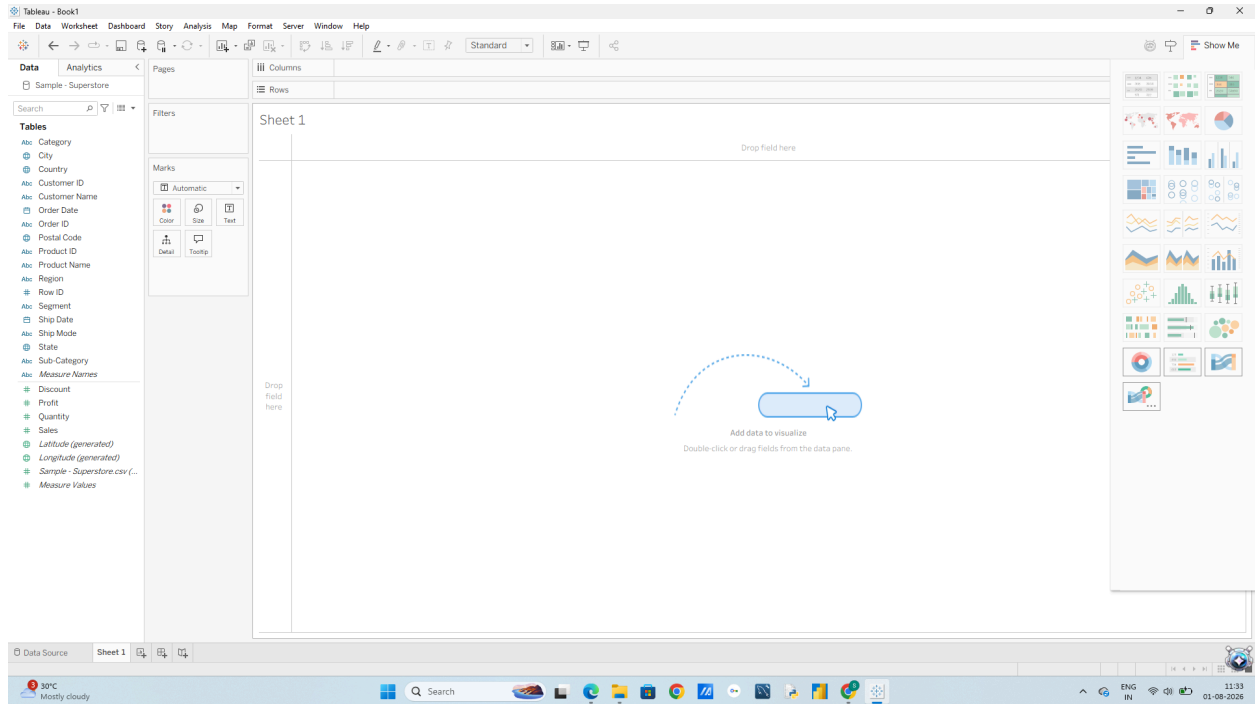
SHETH L.U.J. AND SIR M.V. COLLEGE OF ARTS, SCIENCE AND COMMERCE

Subject: :Subject: Data Visualization with Power BI and Tableau

Practical 1 :1: **Getting Started with Tableau and Building Basic Visualizations**

Shelves (Top/Center): The Columns and Rows shelves act as your X and Y coordinates.

The Marks Card (Left of center): This controls your visual design properties like color, text labels, and sizes.



Dimensions (Categorical / Blue Icons): Fields containing text, dates, or geography. In Superstore, look for fields like Category, Segment, State, or Order Date. They slice your data into distinct group

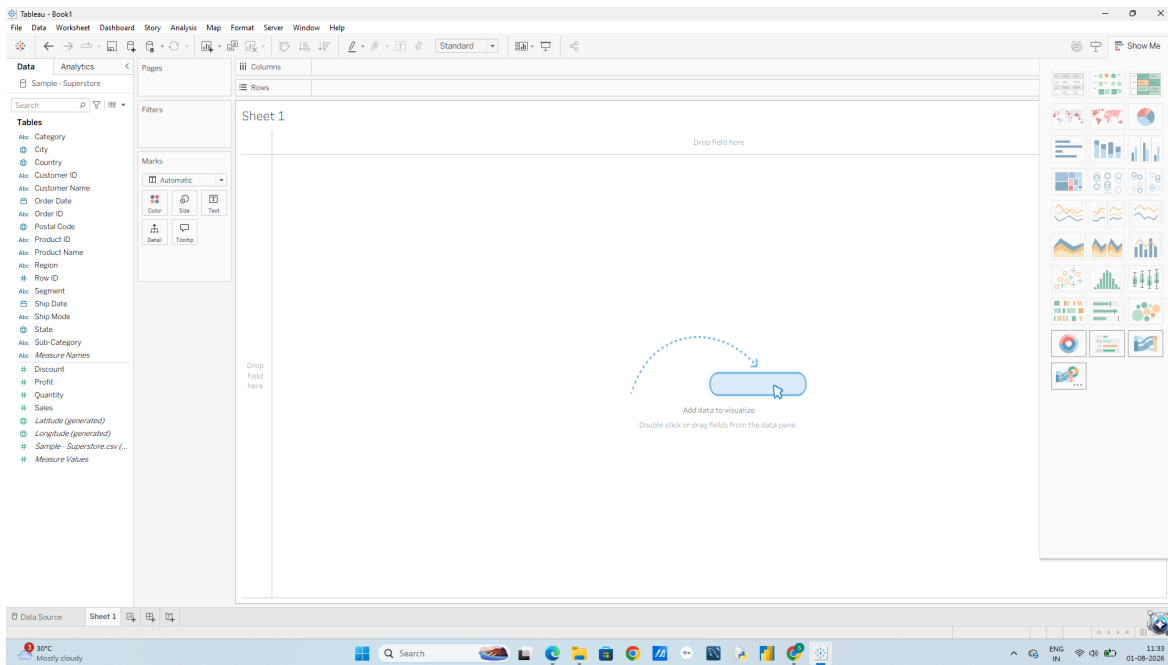
NAME: ADARSH SHUKLA

ROLL NO : T117

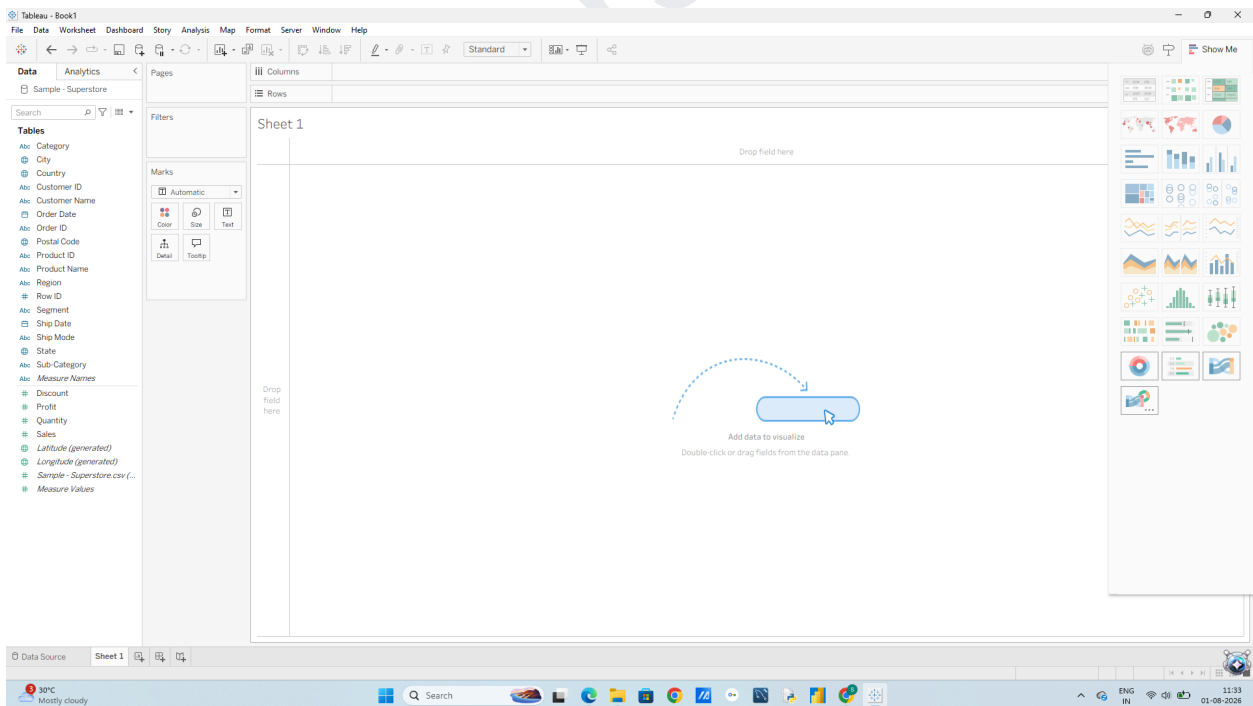
SHETH L.U.J. AND SIR M.V. COLLEGE OF ARTS, SCIENCE AND COMMERCE

Subject: :Subject: Data Visualization with Power BI and Tableau

Practical 1 :1: Getting Started with Tableau and Building Basic Visualizations



Measures (Numerical / Green Icons): Fields containing raw quantitative values. Look for fields like Sales, Profit, or Quantity. Tableau will automatically summarize or average these values whenever you place them on a chart.



Part C: Convert Fields Between Discrete and Continuous Tableau colors its data pills to represent how it visualizes information. Blue means Discrete (separate header blocks) and

NAME: ADARSH SHUKLA

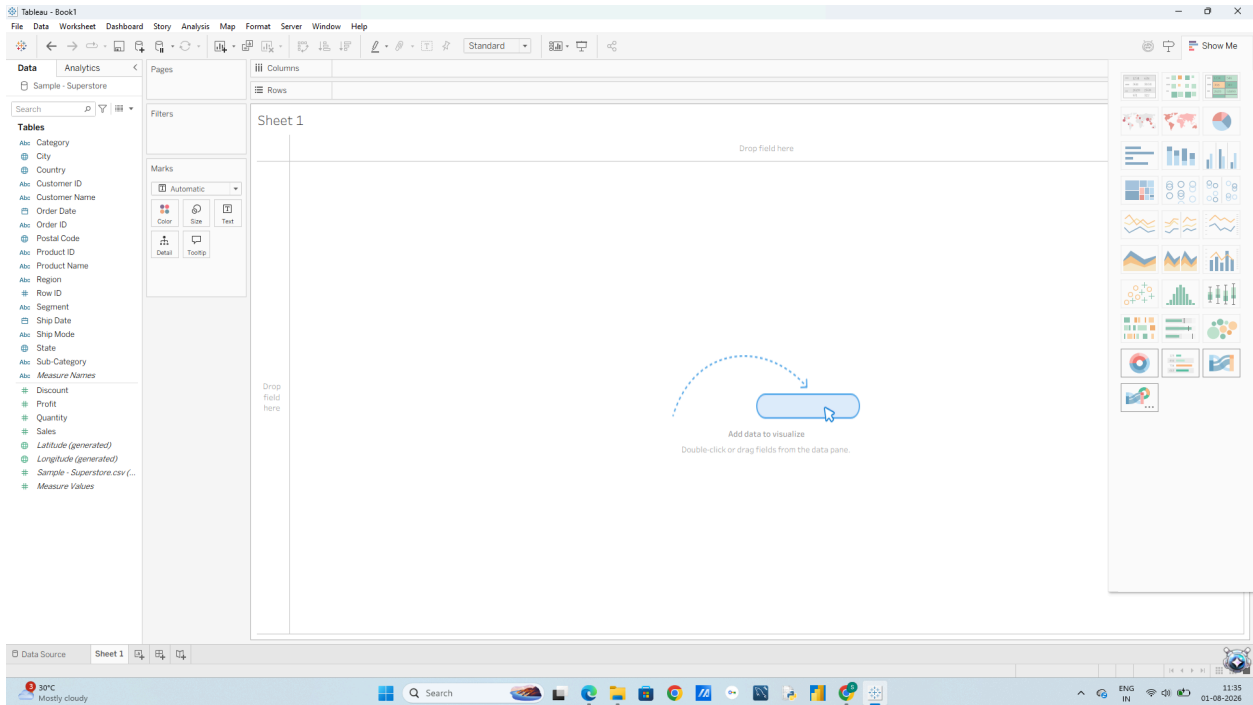
ROLL NO : T117

SHETH L.U.J. AND SIR M.V. COLLEGE OF ARTS, SCIENCE AND COMMERCE

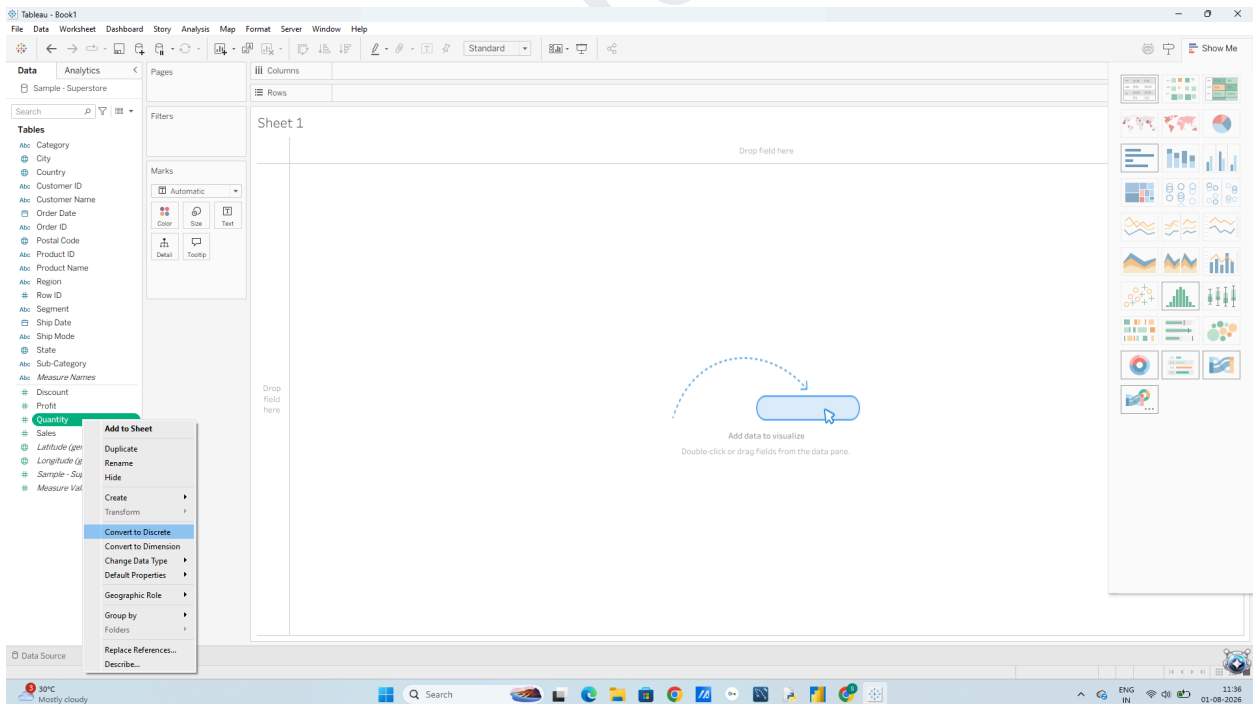
Subject: :Subject: Data Visualization with Power BI and Tableau

Practical 1 :1: **Getting Started with Tableau and Building Basic Visualizations**

Green means Continuous (an infinite unbroken axis). 1. Find the Quantity field in your green measures list



2. Right-click (or hold Ctrl and click) on Quantity and choose Convert to Discrete. Notice its icon changes from a green to a blue.



The Color changes from Green to blue

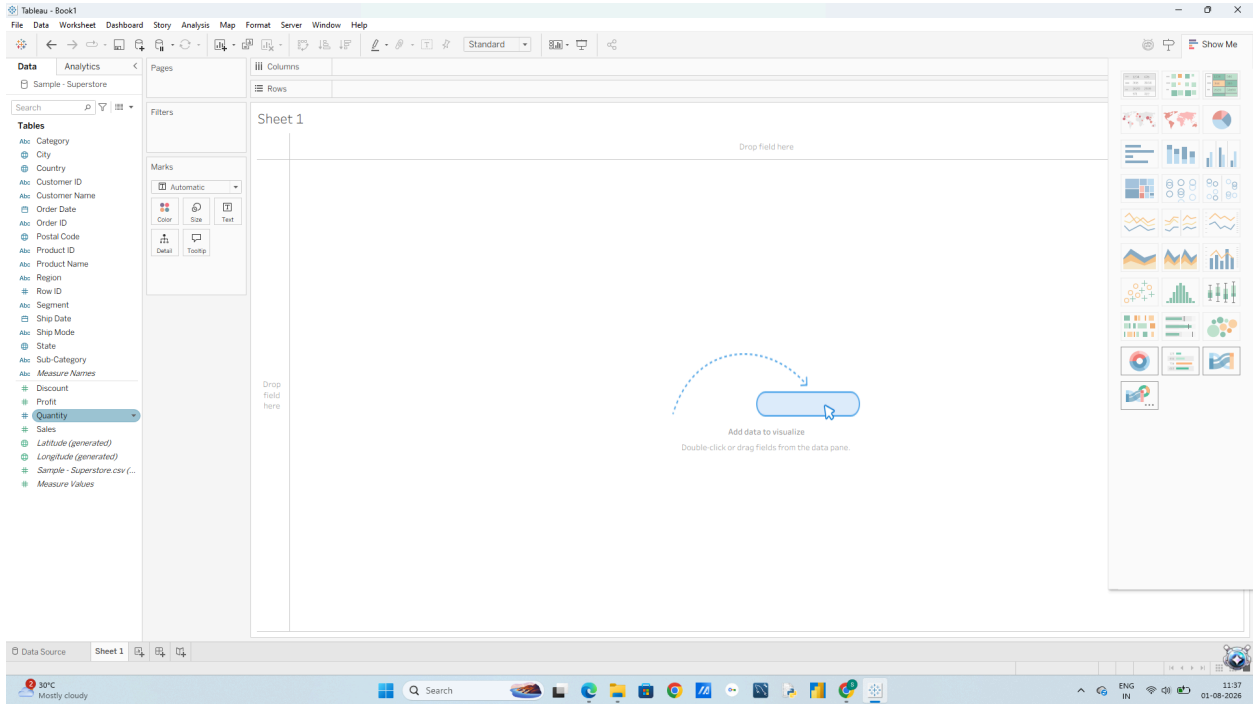
NAME: ADARSH SHUKLA

ROLL NO : T117

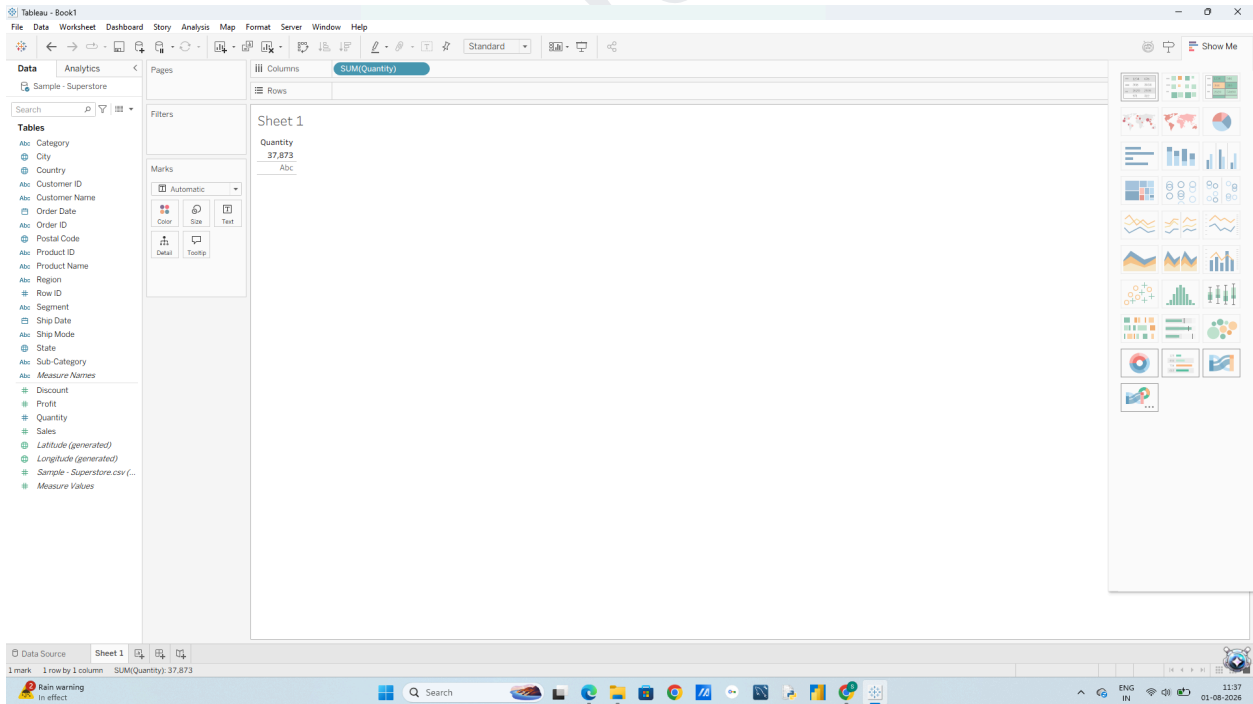
SHETH L.U.J. AND SIR M.V. COLLEGE OF ARTS, SCIENCE AND COMMERCE

Subject: :Subject: Data Visualization with Power BI and Tableau

Practical 1 :1: Getting Started with Tableau and Building Basic Visualizations



3. Drag this new blue Quantity pill onto your Columns shelf. It will draw isolated individual header columns for each exact number (1, 2, 3, etc.).



NAME: ADARSH SHUKLA

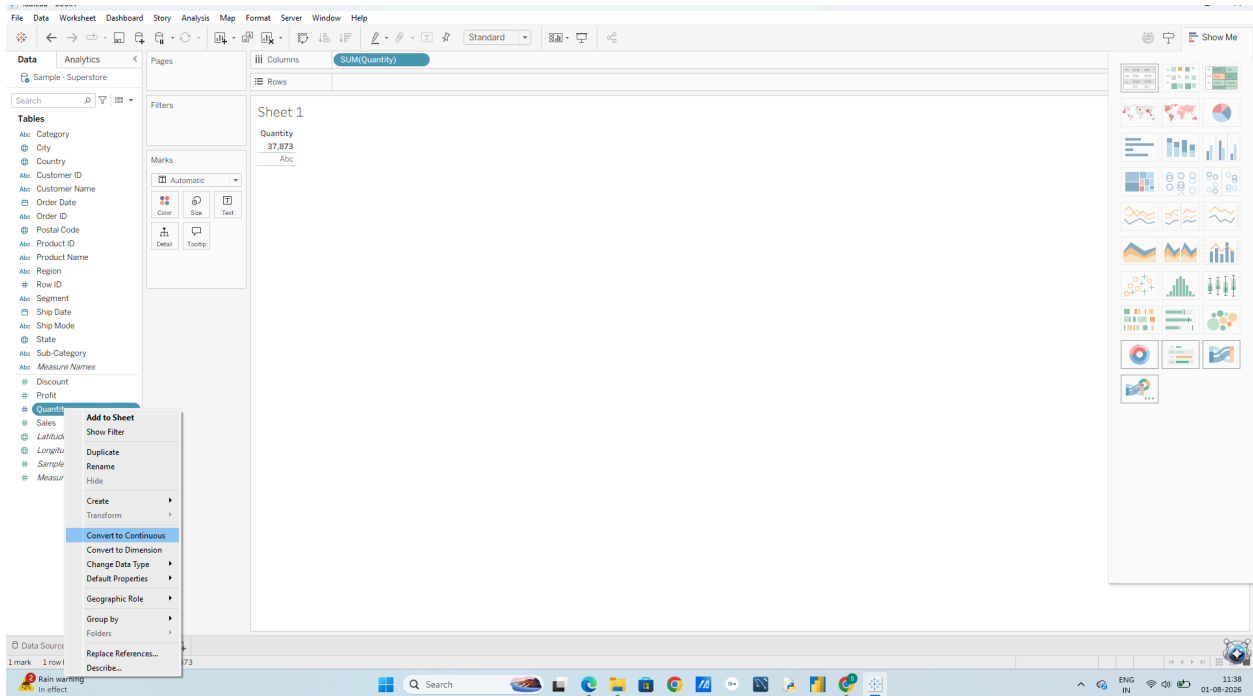
ROLL NO : T117

SHETH L.U.J. AND SIR M.V. COLLEGE OF ARTS, SCIENCE AND COMMERCE

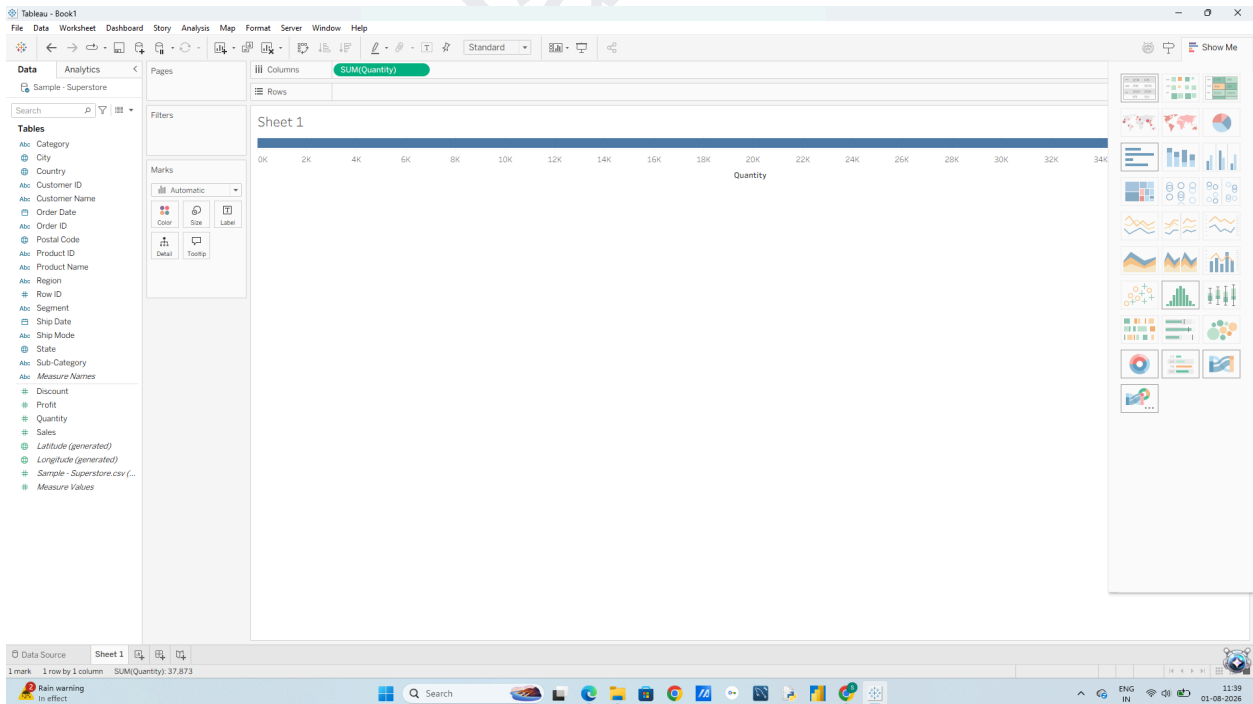
Subject: :Subject: Data Visualization with Power BI and Tableau

Practical 1 :1: **Getting Started with Tableau and Building Basic Visualizations**

4. Drag it back off the shelf. Right-click **Quantity** again and select **Convert to Continuous**



5. Drag it back onto Columns. Instead of individual column blocks, Tableau now draws a singular, smooth, continuous numeric axis line.



NAME: ADARSH SHUKLA
ROLL NO : T117

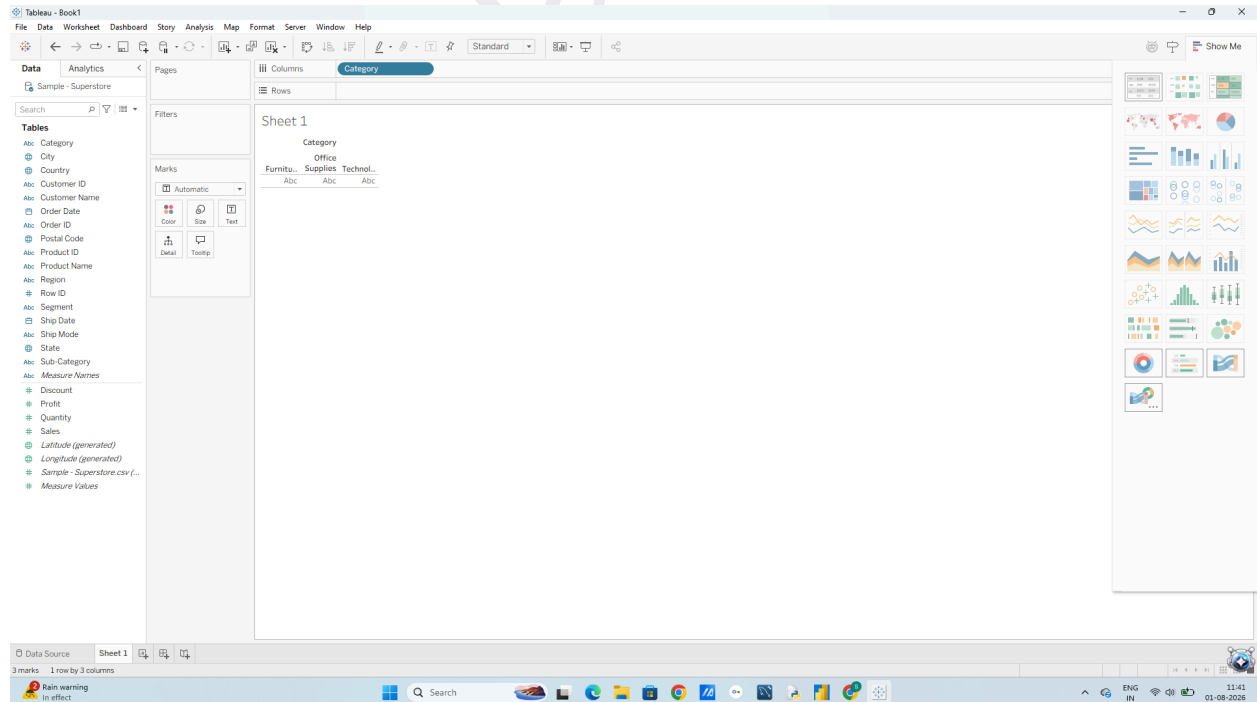
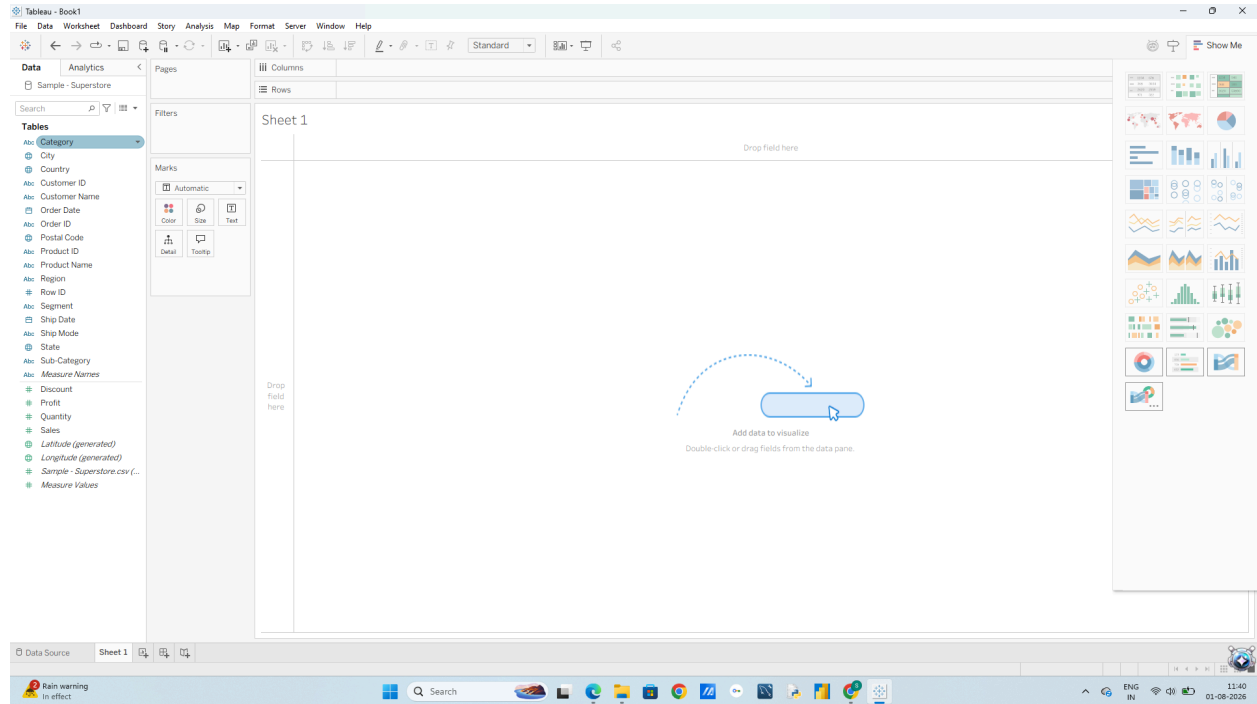
SHETH L.U.J. AND SIR M.V. COLLEGE OF ARTS, SCIENCE AND COMMERCE

Subject: :Subject: Data Visualization with Power BI and Tableau

Practical 1 :1: Getting Started with Tableau and Building Basic Visualizations

Part D: Create and Customize a Basic Bar Chart

1. From your Data Pane, find the Category dimension and drag it up onto the Columns shelf

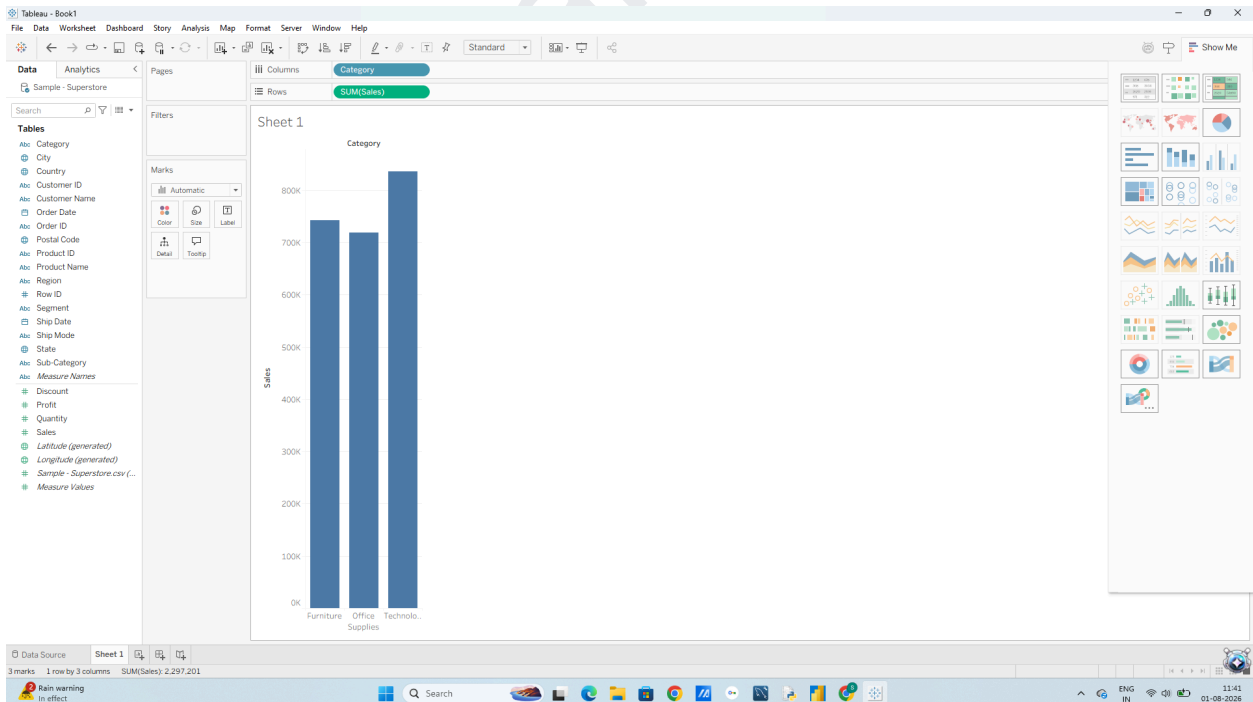
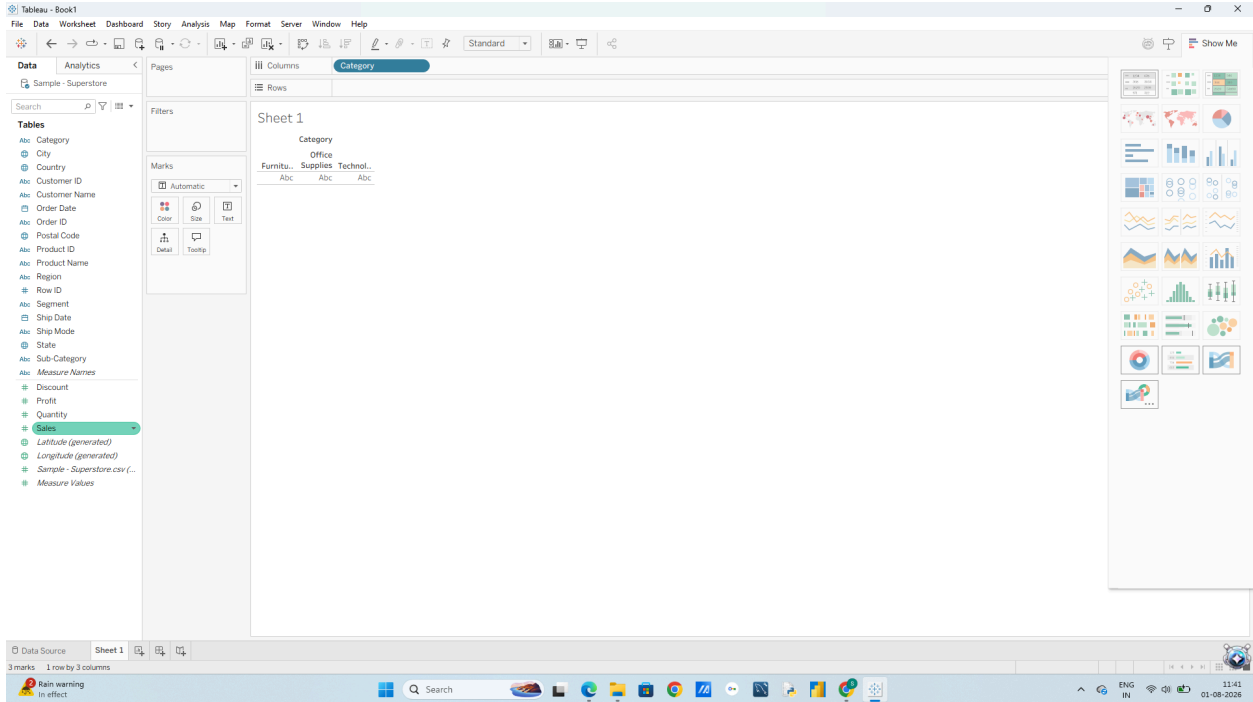


NAME: ADARSH SHUKLA

ROLL NO : T117

Practical 1 :1: **Getting Started with Tableau and Building Basic Visualizations**

2. Find the Sales measure and drag it up onto the Rows shelf. You will instantly get three vertical bars representing Furniture, Office Supplies, and Technology.

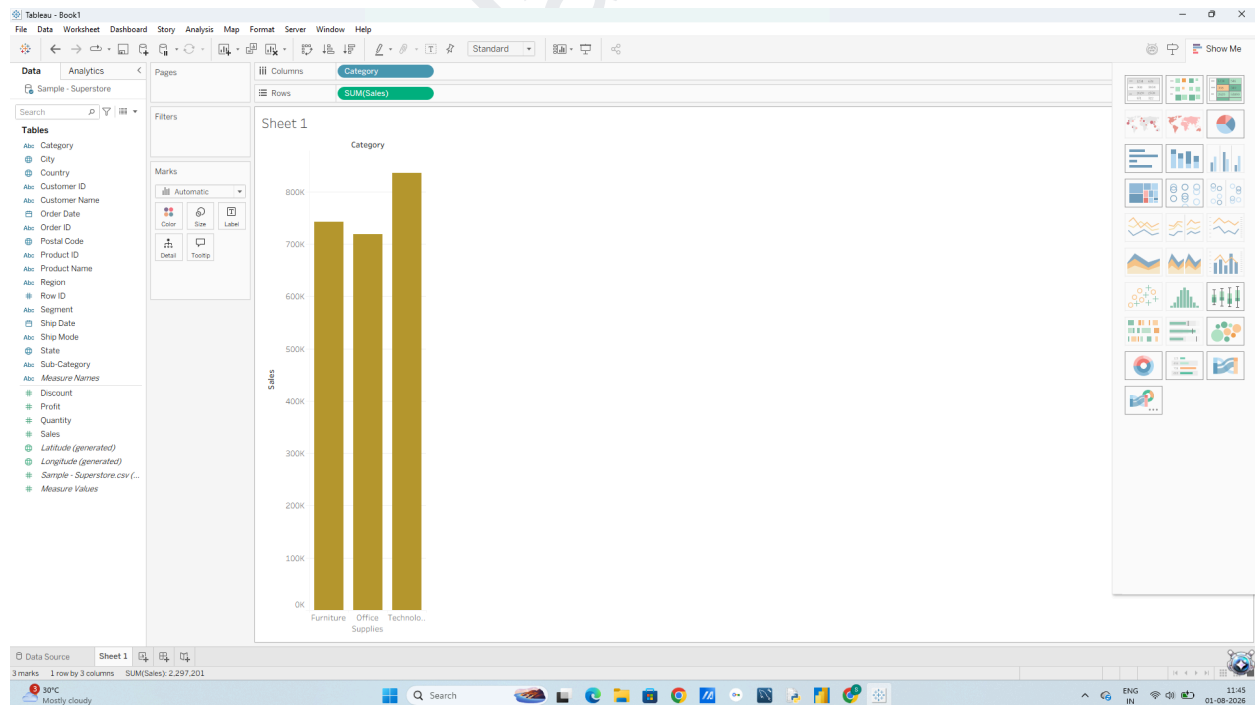
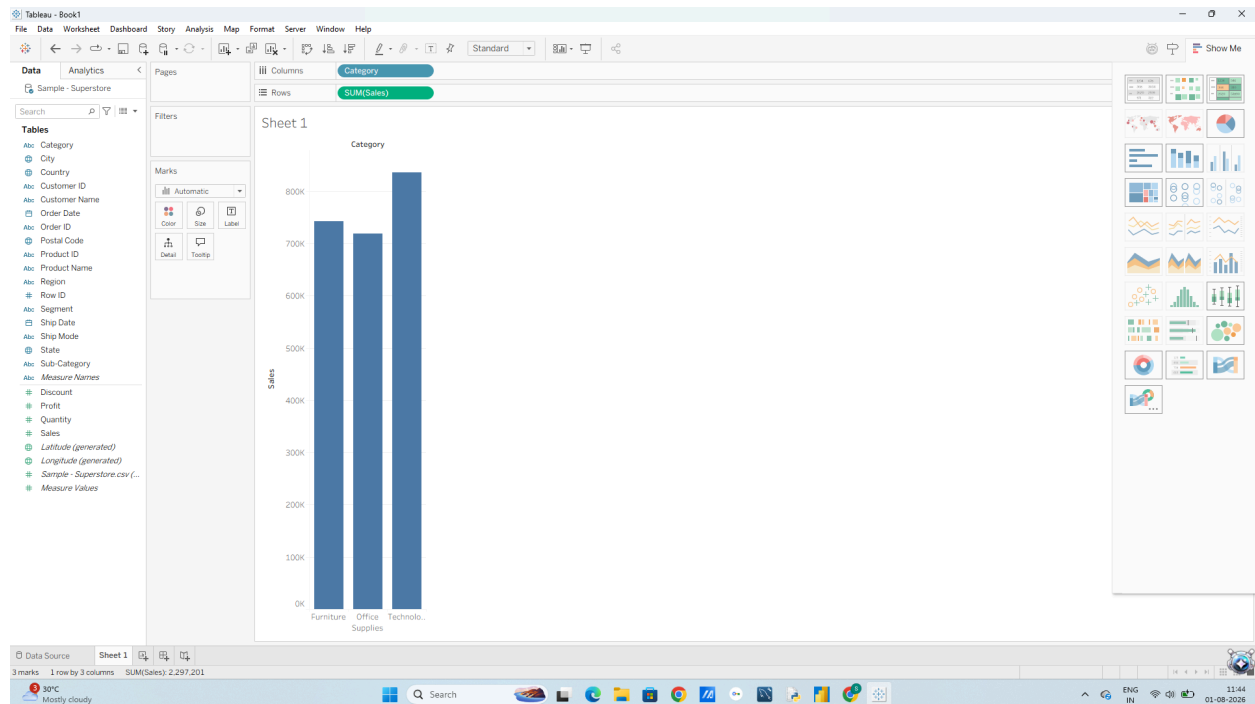


SHETH L.U.J. AND SIR M.V. COLLEGE OF ARTS, SCIENCE AND COMMERCE

Subject: :Subject: Data Visualization with Power BI and Tableau

Practical 1 :1: Getting Started with Tableau and Building Basic Visualizations

3. Color customization: Take Category from the Data pane a second time, and drop it directly onto the Color tile in the Marks card.



NAME: ADARSH SHUKLA

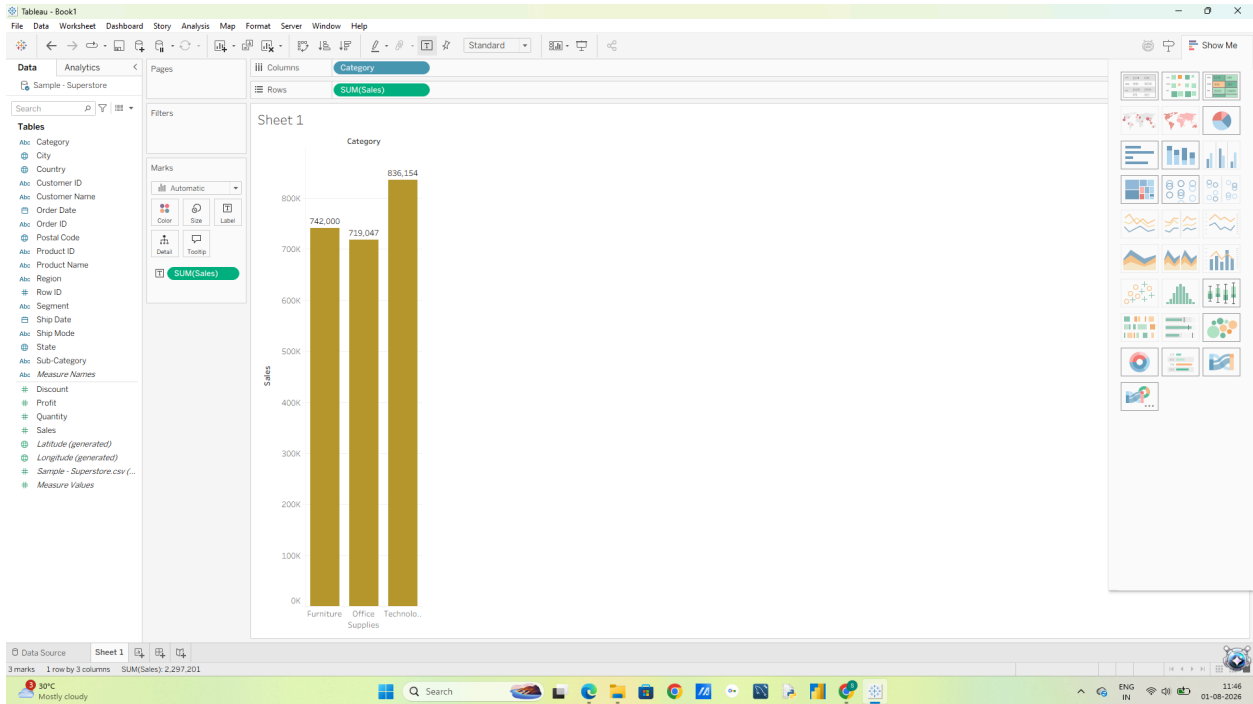
ROLL NO : T117

SHETH L.U.J. AND SIR M.V. COLLEGE OF ARTS, SCIENCE AND COMMERCE

Subject: :Subject: Data Visualization with Power BI and Tableau

Practical 1 :1: **Getting Started with Tableau and Building Basic Visualizations**

4. Label customization: Drag the Sales pill from the data pane and drop it onto the Label tile in the Marks card to display the precise revenue numbers over the bars

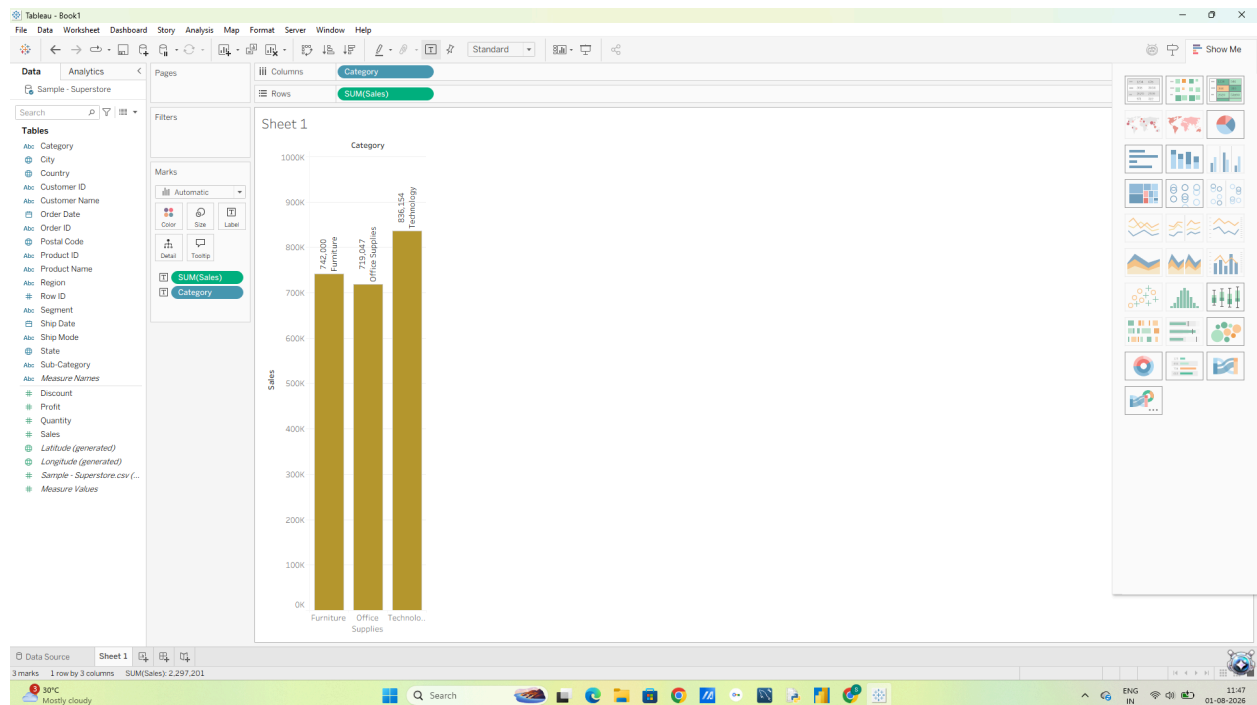


NAME: ADARSH SHUKLA
ROLL NO : T117

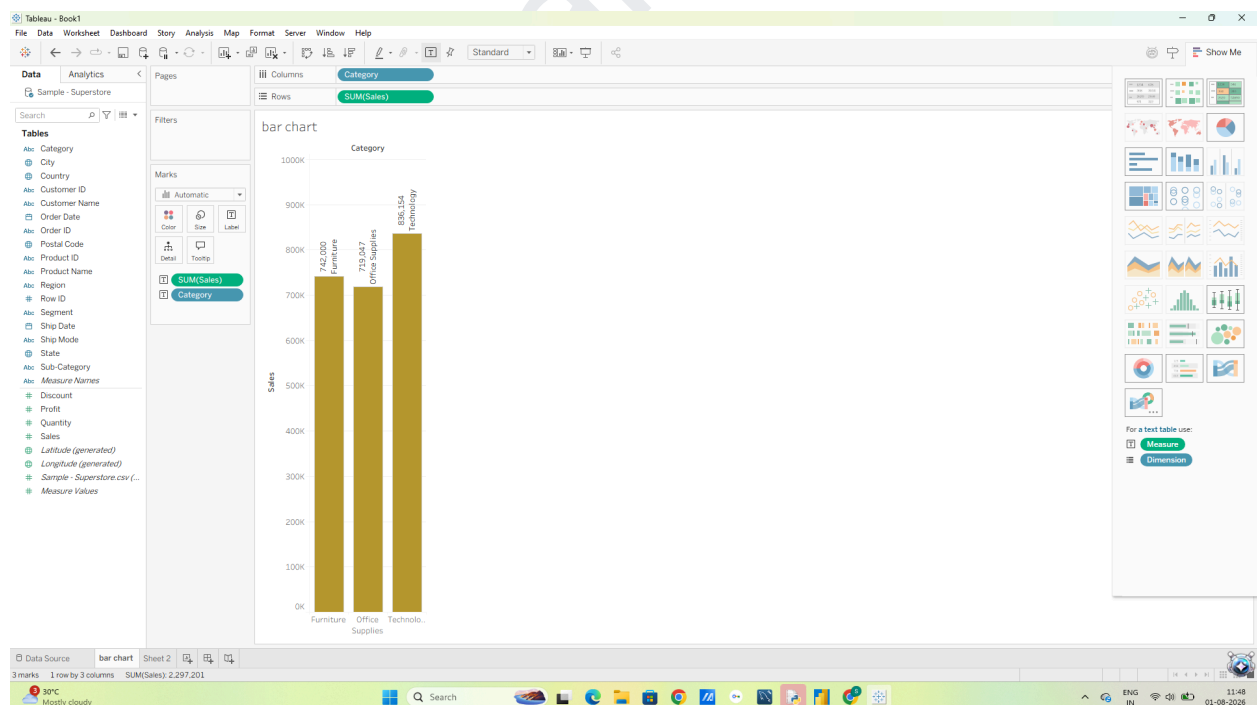
SHETH L.U.J. AND SIR M.V. COLLEGE OF ARTS, SCIENCE AND COMMERCE

Subject: :Subject: Data Visualization with Power BI and Tableau

Practical 1 :1: Getting Started with Tableau and Building Basic Visualizations



5. Double-click the bottom tab sheet name Sheet 1 and rename it to Category Bar Chart



NAME: ADARSH SHUKLA
ROLL NO : T117

SHETH L.U.J. AND SIR M.V. COLLEGE OF ARTS, SCIENCE AND COMMERCE

Subject: :Subject: Data Visualization with Power BI and Tableau

Practical 1 :1: **Getting Started with Tableau and Building Basic Visualizations**

Part E: Build a Line Chart for Time-Series Analysis 1. Click the New Worksheet button at the very bottom tab bar (the small icon with a chart and a plus symbol).

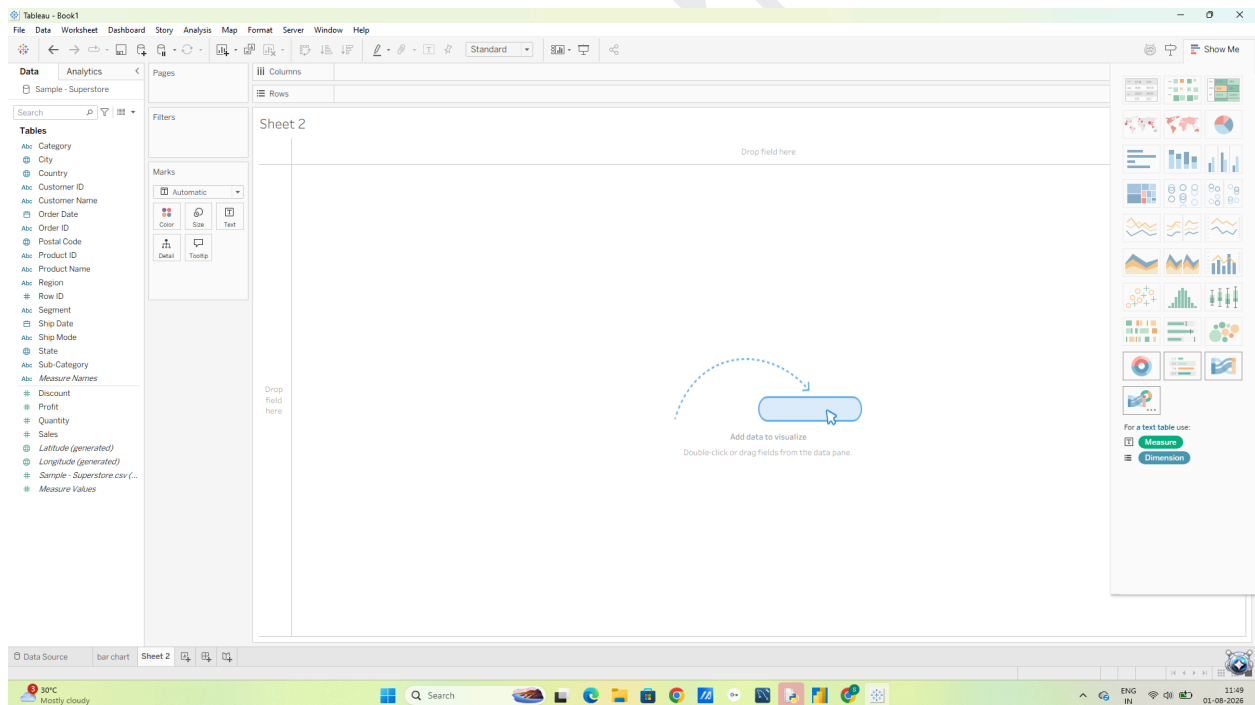
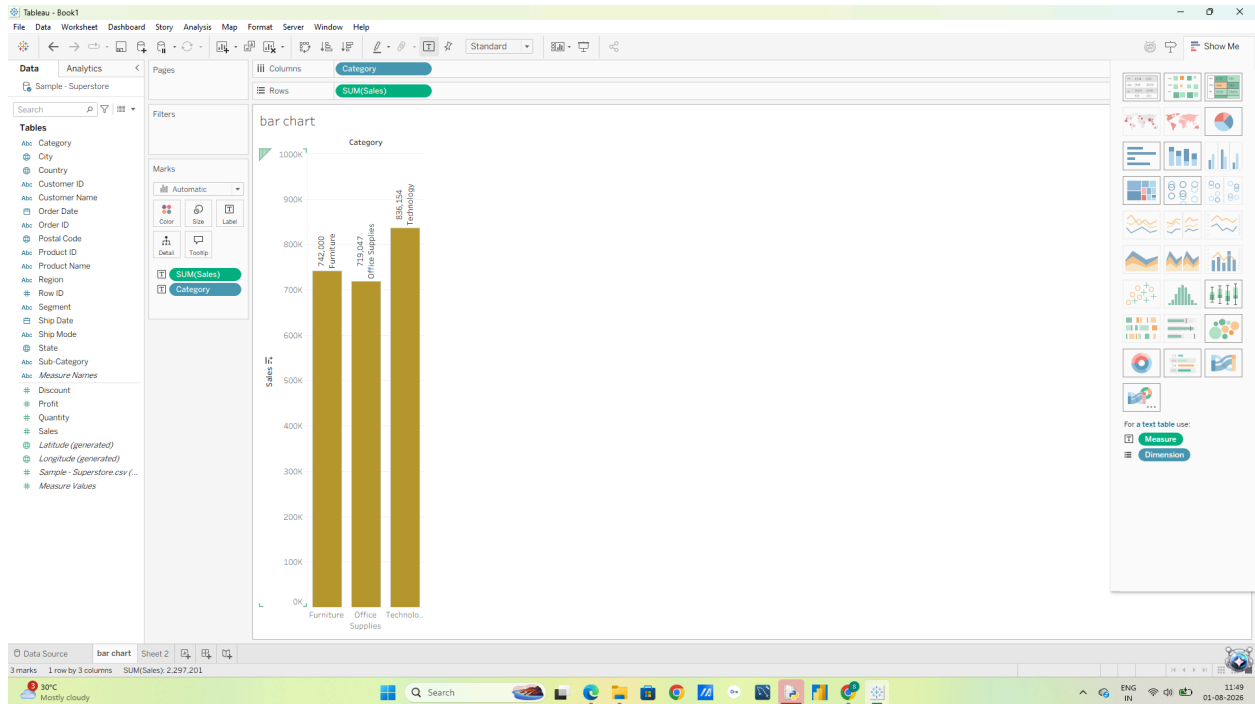
NAME: ADARSH SHUKLA

ROLL NO : T117

SHETH L.U.J. AND SIR M.V. COLLEGE OF ARTS, SCIENCE AND COMMERCE

Subject: :Subject: Data Visualization with Power BI and Tableau

Practical 1 :1: Getting Started with Tableau and Building Basic Visualizations



2. Drag the Order Date dimension onto your Columns shelf.

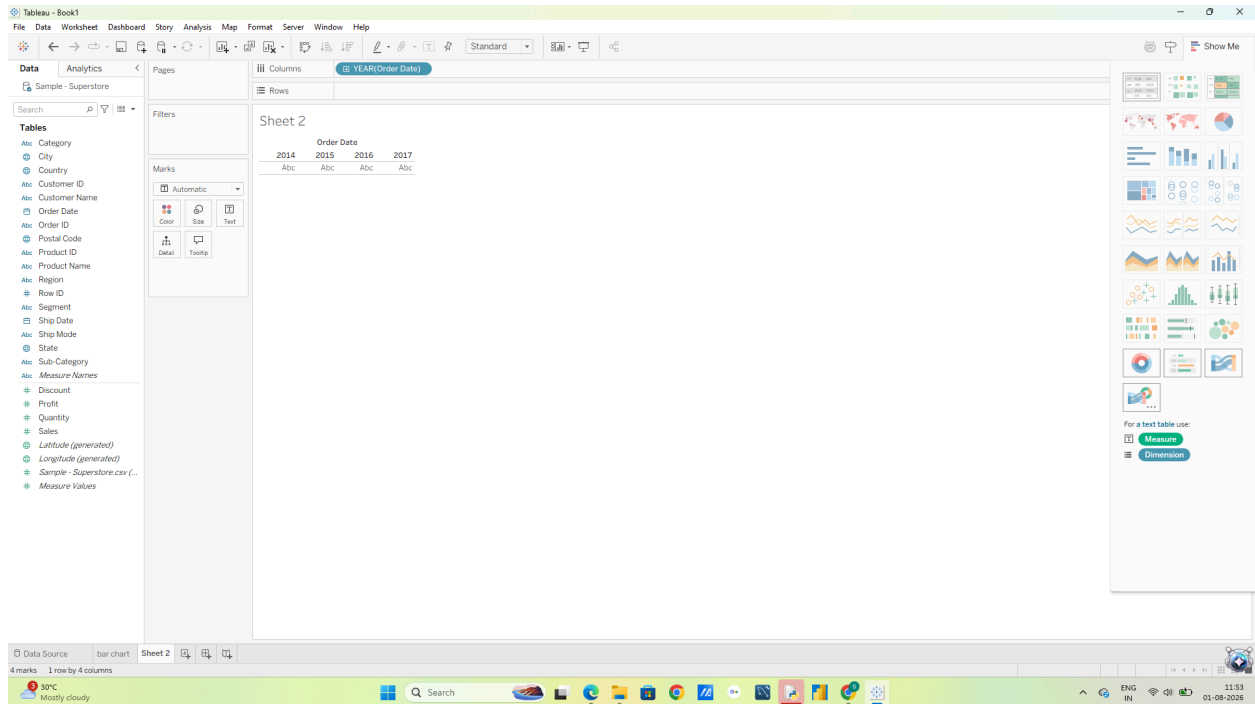
NAME: ADARSH SHUKLA

ROLL NO : T117

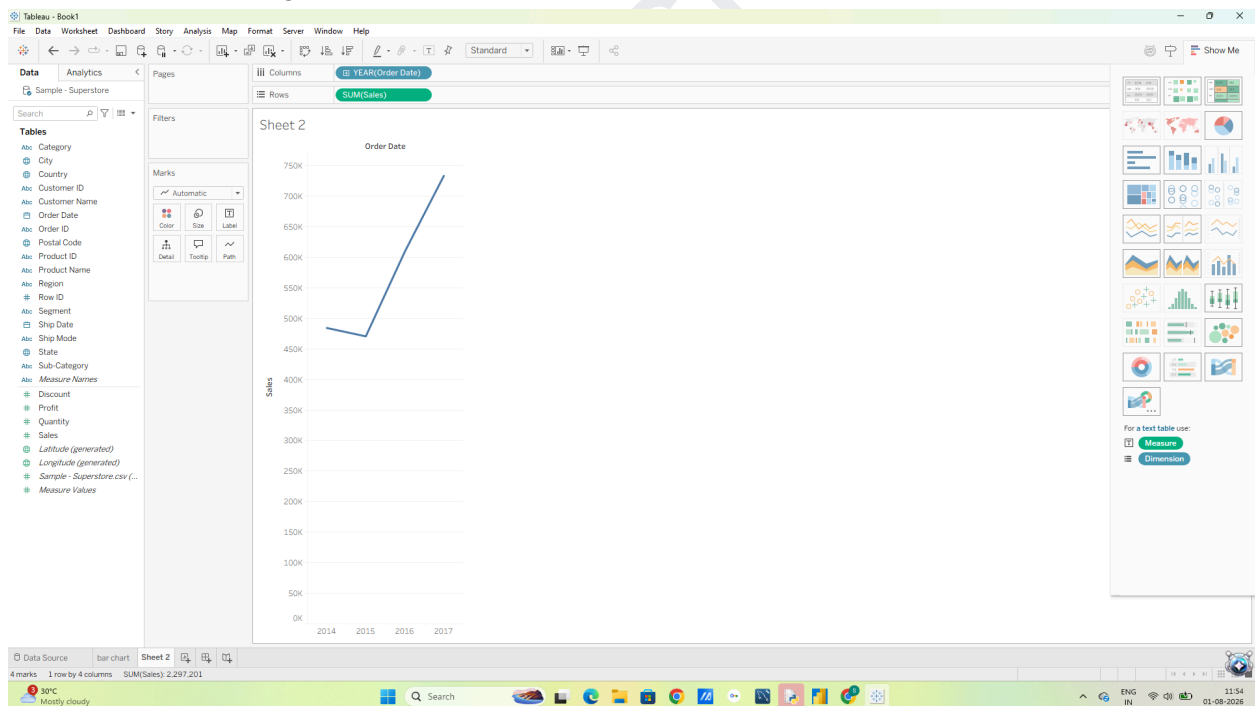
SHETH L.U.J. AND SIR M.V. COLLEGE OF ARTS, SCIENCE AND COMMERCE

Subject: :Subject: Data Visualization with Power BI and Tableau

Practical 1 :1: Getting Started with Tableau and Building Basic Visualizations



3. Drag the Sales measure onto your Rows shelf. Because a date field is on your columns shelf, Tableau defaults straight into a line chart.



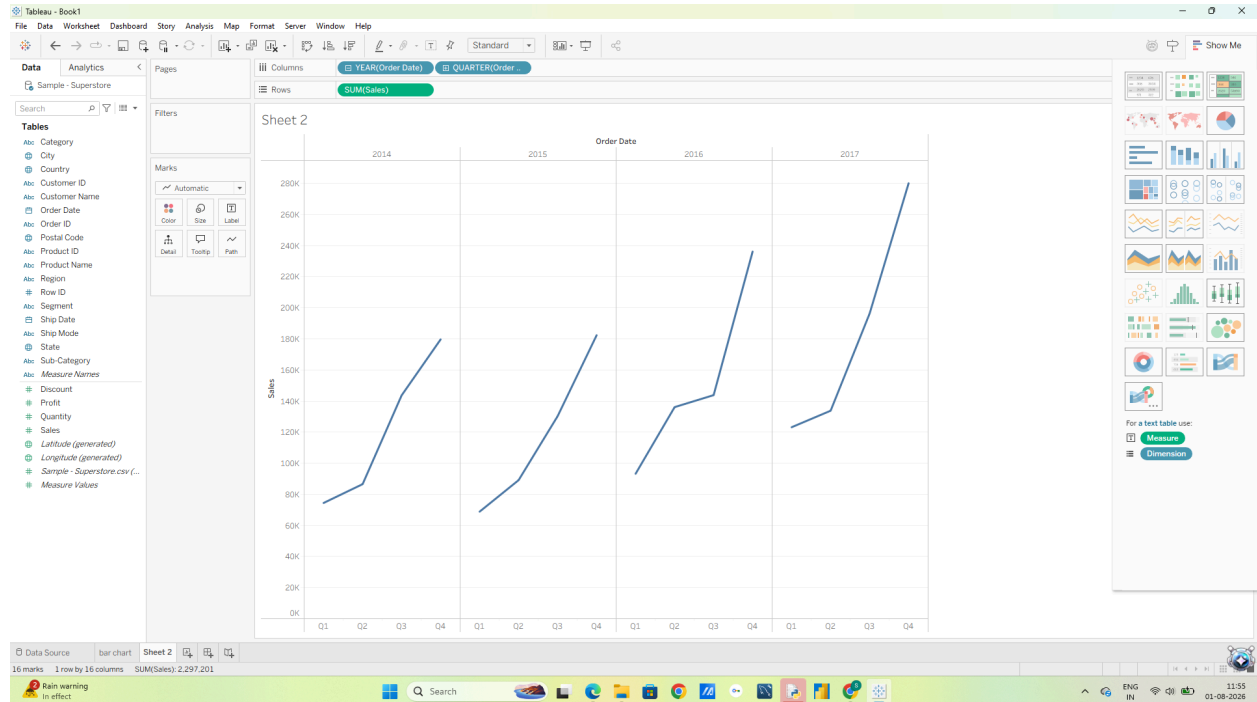
NAME: ADARSH SHUKLA
ROLL NO : T117

SHETH L.U.J. AND SIR M.V. COLLEGE OF ARTS, SCIENCE AND COMMERCE

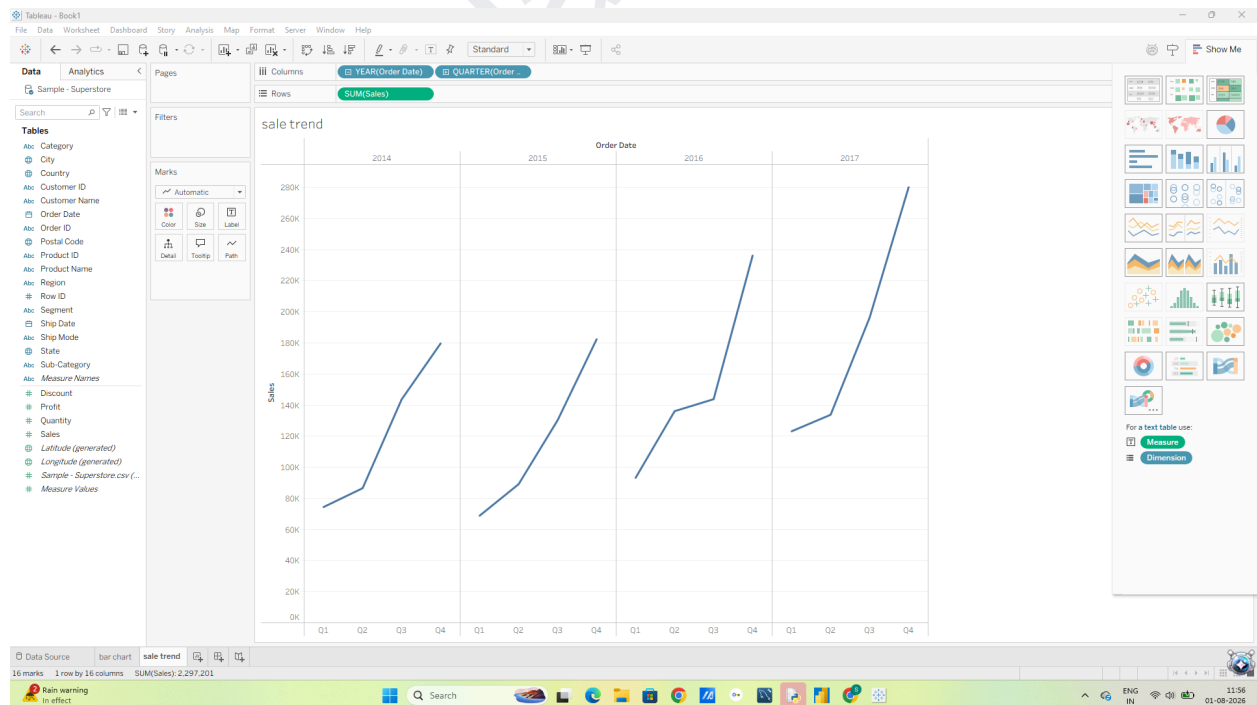
Subject: :Subject: Data Visualization with Power BI and Tableau

Practical 1 :1: **Getting Started with Tableau and Building Basic Visualizations**

4. Drilling Down: Look at the blue YEAR(Order Date) pill sitting in your Columns shelf. Click the small + icon located on that blue pill. It will expand your data into Quarters and Months, revealing hidden sales trends over time.

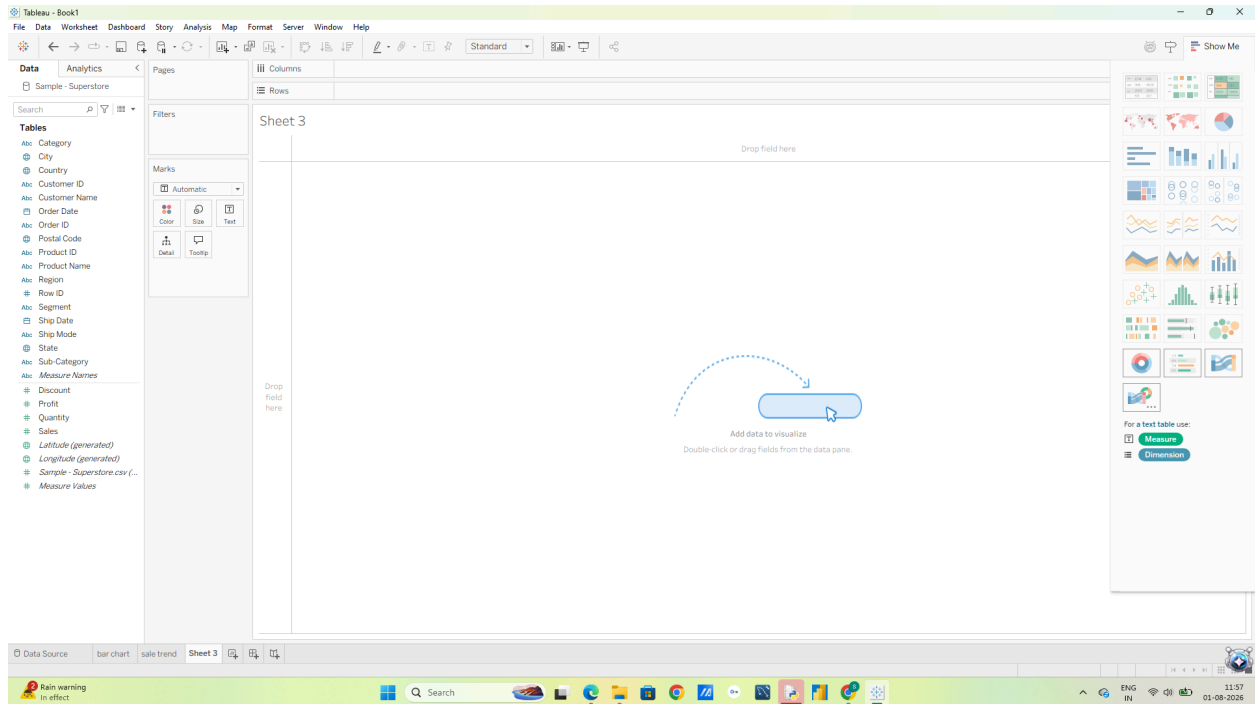


5. Double-click the sheet tab at the bottom and rename this worksheet to Sales Trend



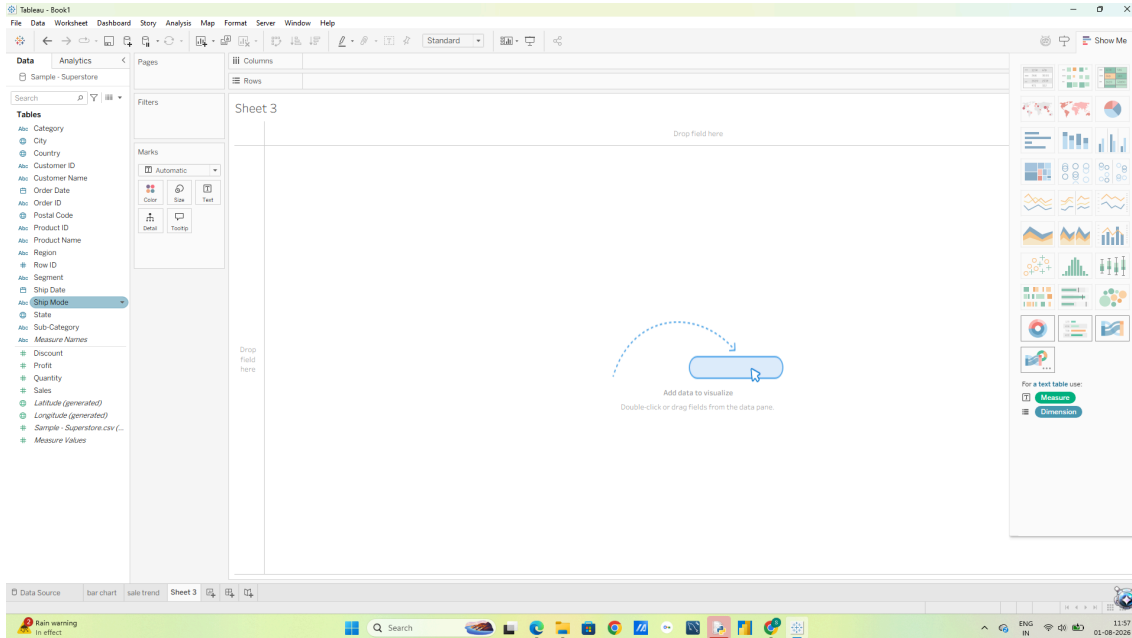
NAME: ADARSH SHUKLA
ROLL NO : T117

Part F: Create a Geographic Map Visualization 1. Click the New Worksheet button at the bottom to create your third sheet.

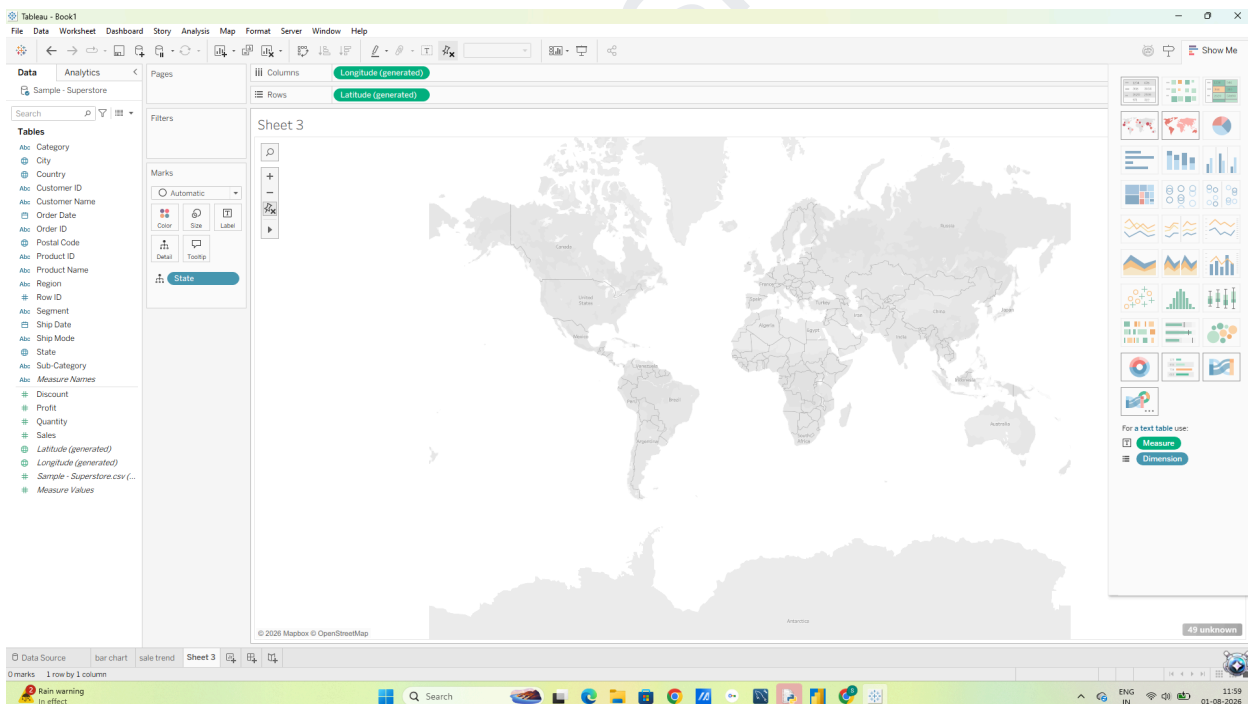


2. Look through your Data Pane and find the State field which is marked in red circle. It should display a tiny globe icon next to it, indicating geographic data.

Practical 1 :1: **Getting Started with Tableau and Building Basic Visualizations**



3. Double-click the State field. Tableau will automatically inject Latitude and Longitude coordinates into your shelves and draw a map.



4. Styling the Map: Go to your Data Pane, pick up Profit, and drop it directly onto the Color icon inside your Marks card. This will instantly color-code each state based on its performance

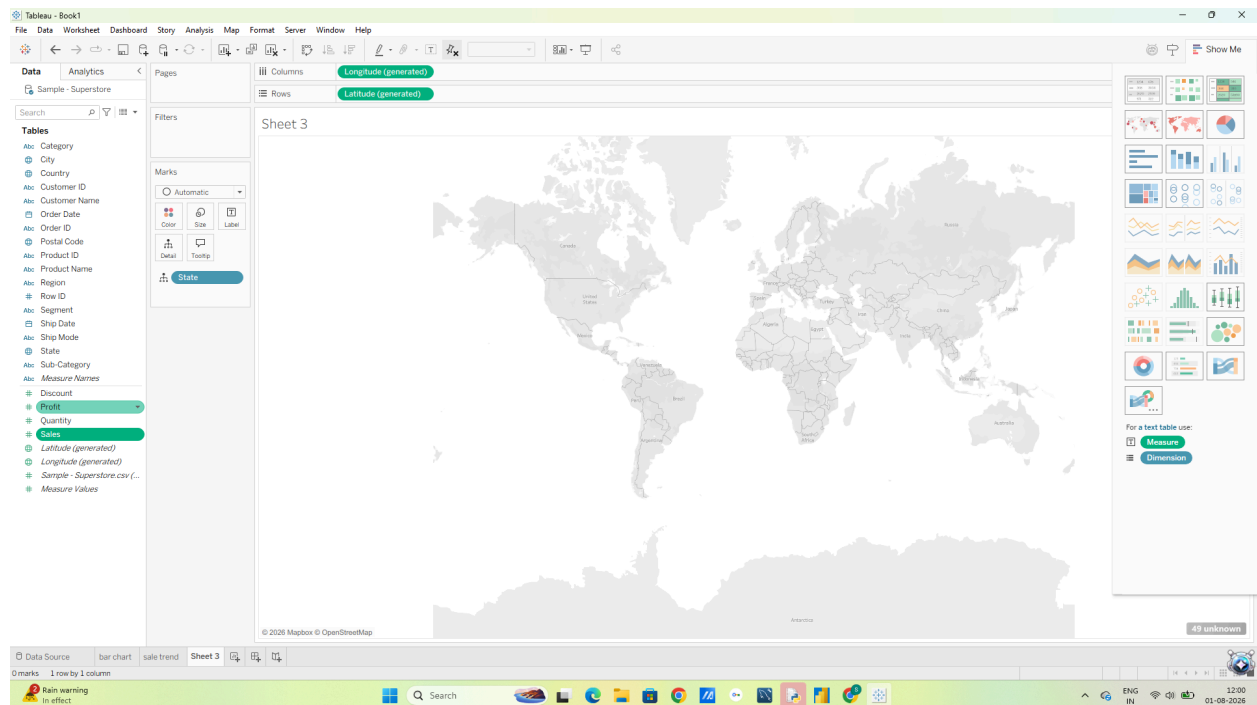
NAME: ADARSH SHUKLA

ROLL NO : T117

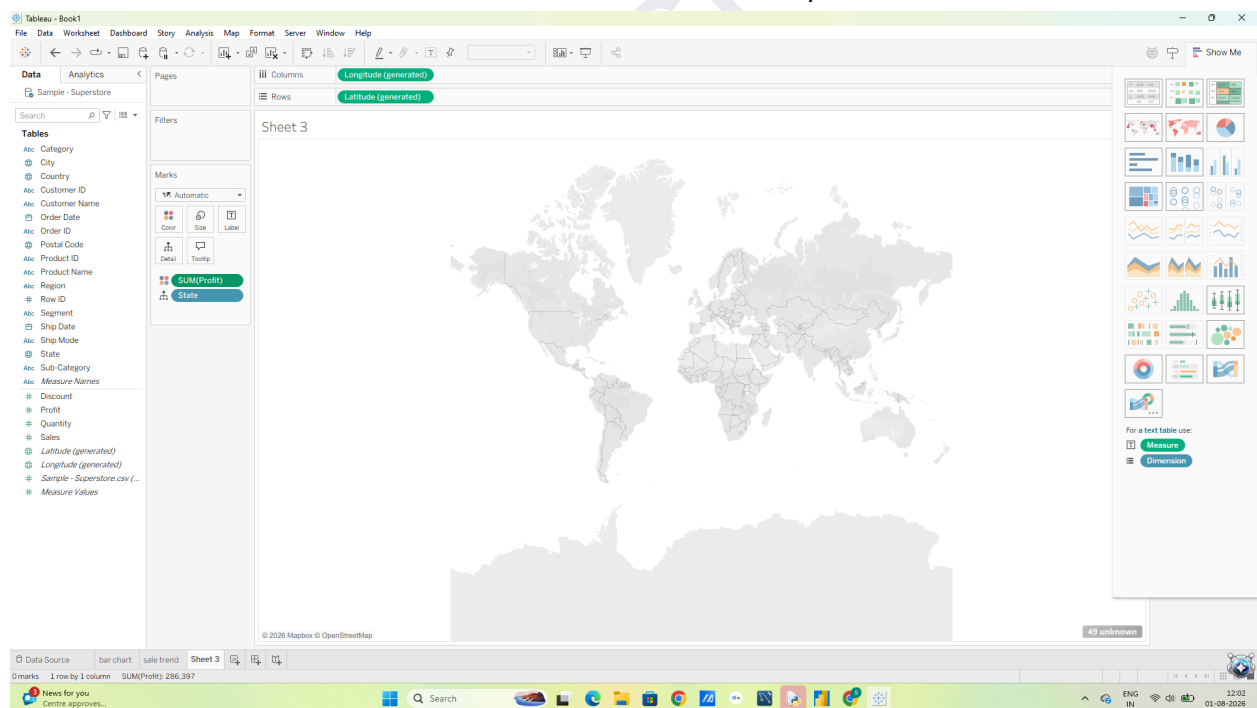
SHETH L.U.J. AND SIR M.V. COLLEGE OF ARTS, SCIENCE AND COMMERCE

Subject: :Subject: Data Visualization with Power BI and Tableau

Practical 1 :1: Getting Started with Tableau and Building Basic Visualizations



5. Rename this worksheet tab at the bottom to State Profit Map.



Part G: Combine Charts into an Interactive Dashboard 1. Look at the bottom navigation tab bar and click the New Dashboard icon (it looks like a small 2x2 grid window pane with a plus symbol).

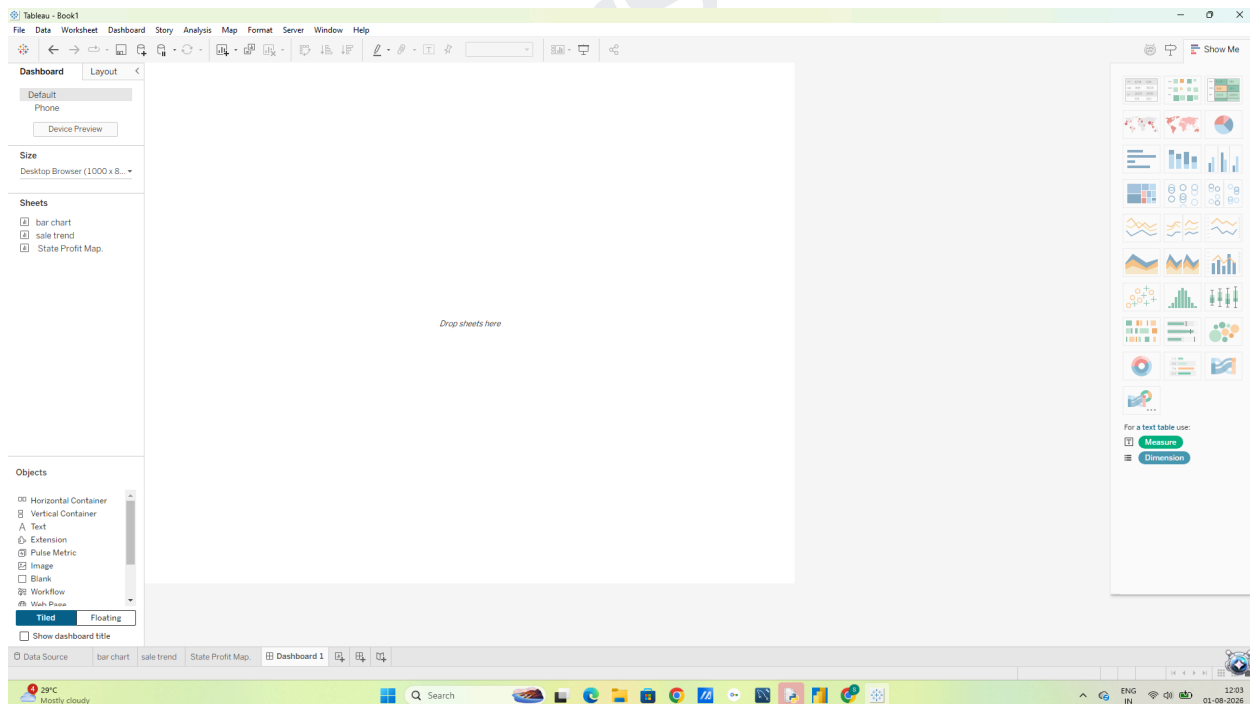
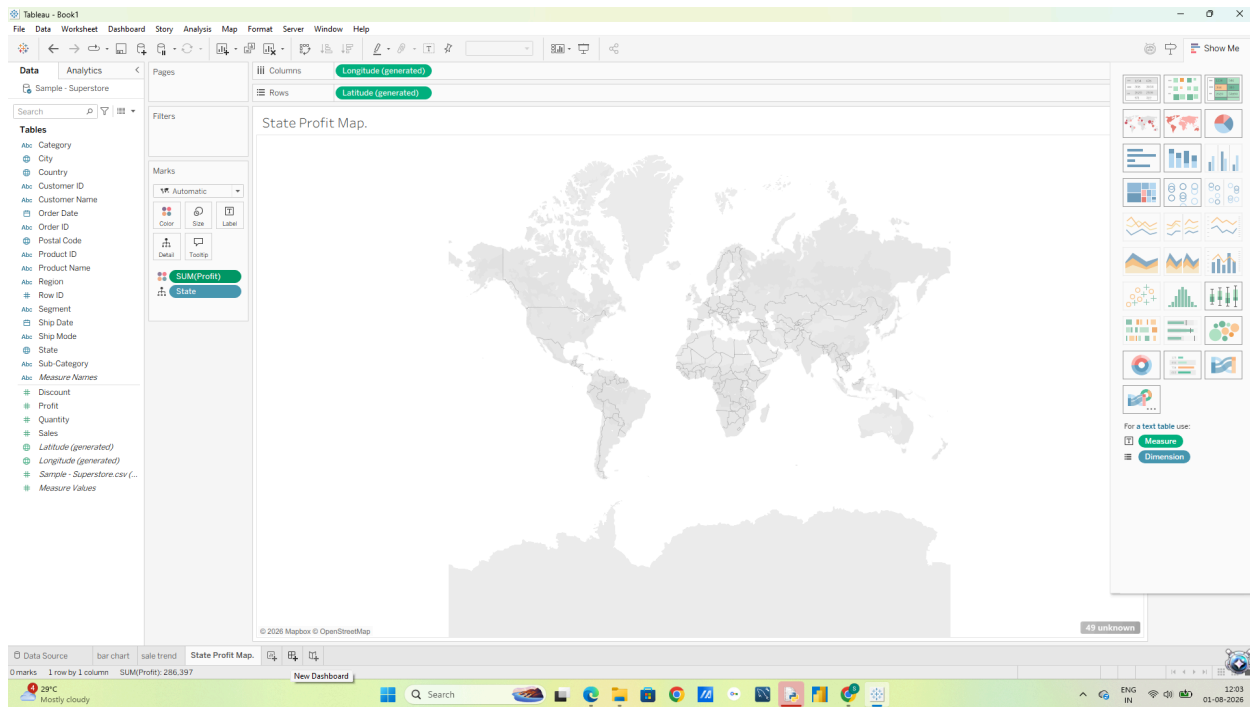
NAME: ADARSH SHUKLA

ROLL NO : T117

SHETH L.U.J. AND SIR M.V. COLLEGE OF ARTS, SCIENCE AND COMMERCE

Subject: :Subject: Data Visualization with Power BI and Tableau

Practical 1 :1: Getting Started with Tableau and Building Basic Visualizations

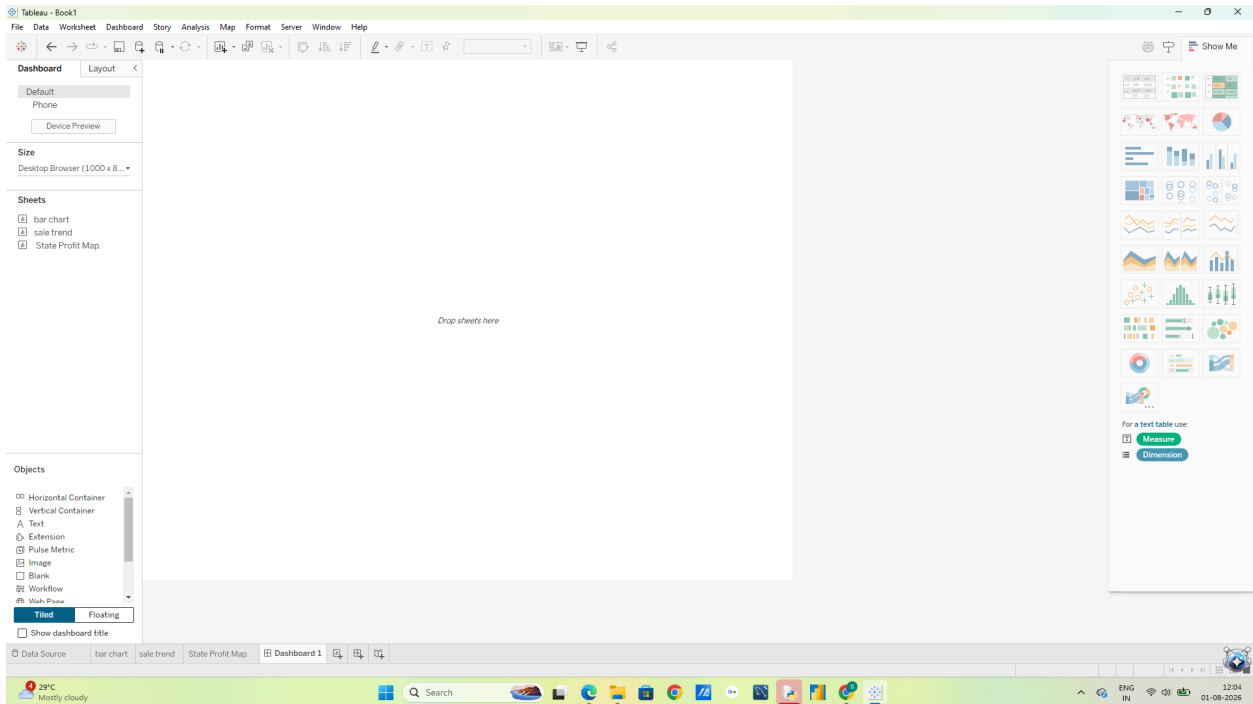


2. On the left side panel, you will see a list of your worksheets: Category Bar Chart, Sales Trend, and State Profit Map

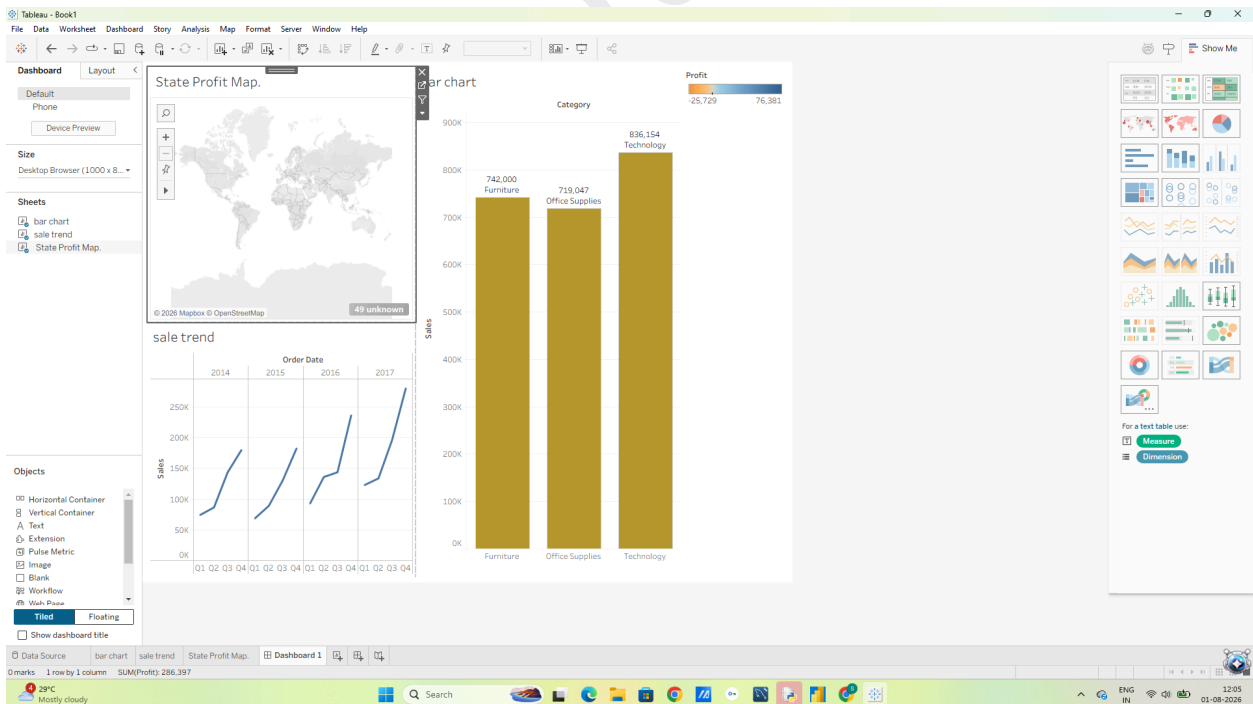
NAME: ADARSH SHUKLA

ROLL NO : T117

Practical 1 :1: Getting Started with Tableau and Building Basic Visualizations



3. Drag your Category Bar Chart sheet onto the empty dashboard canvas area.

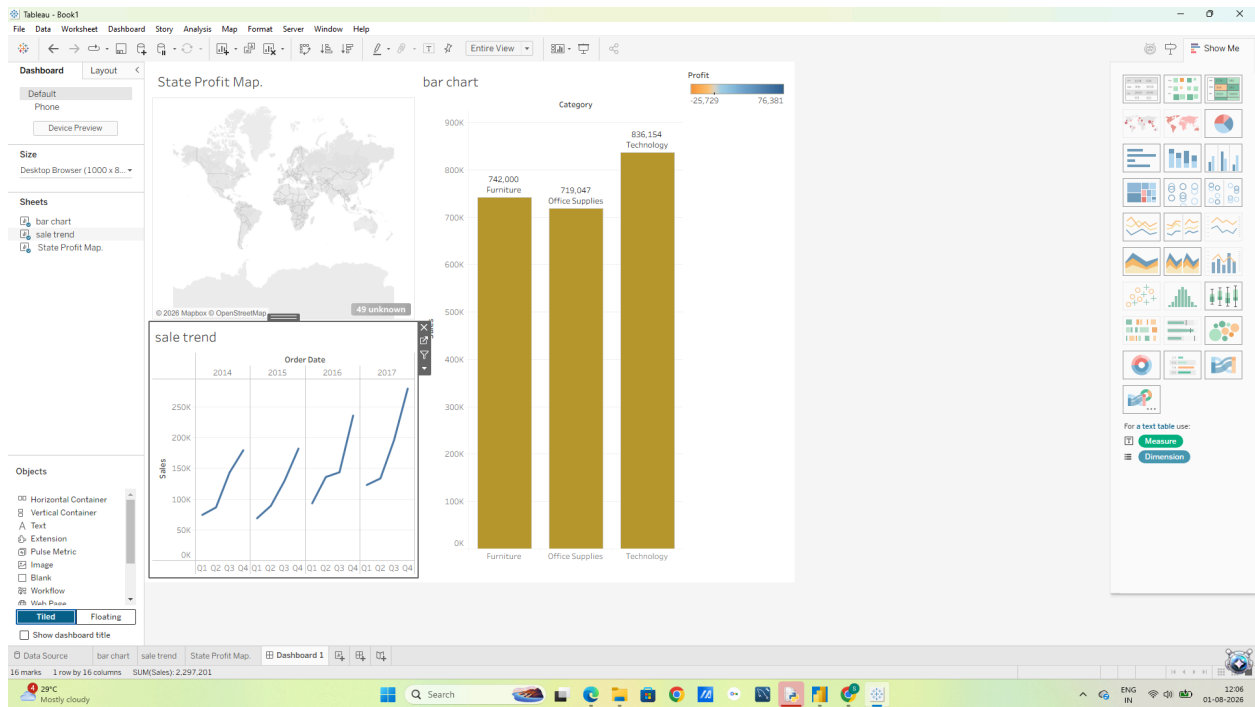


NAME: ADARSH SHUKLA

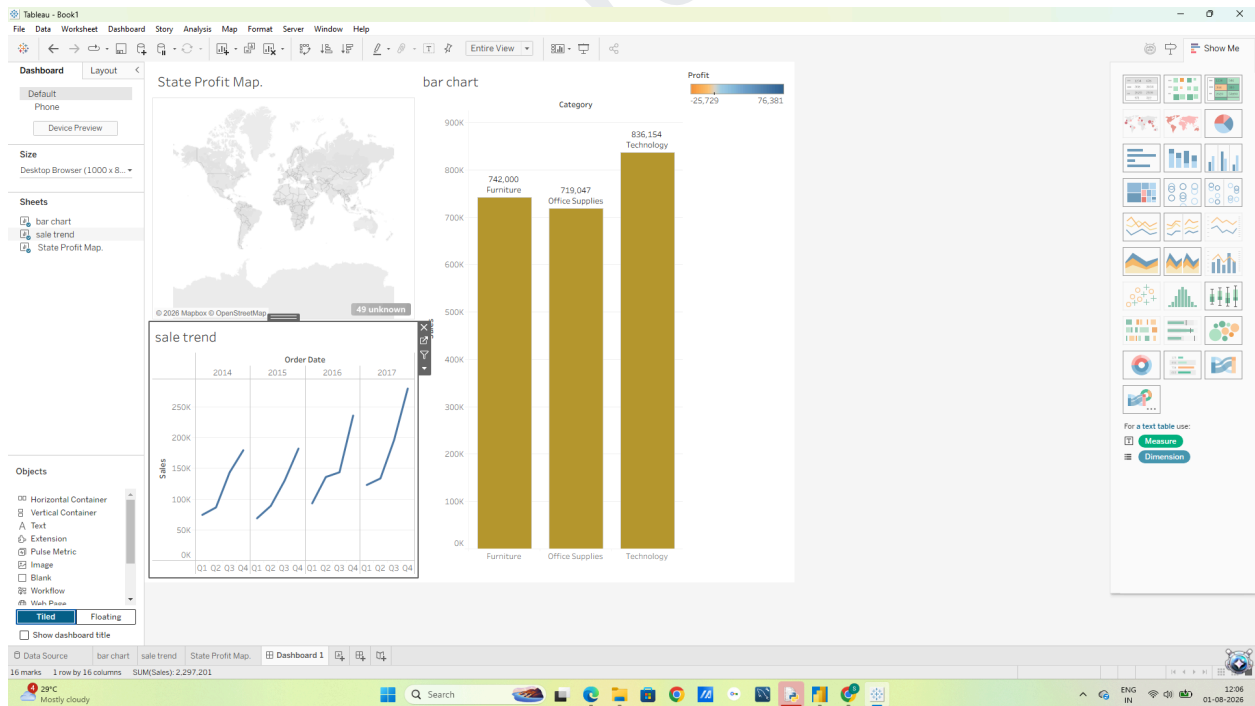
ROLL NO : T117

Practical 1 :1: **Getting Started with Tableau and Building Basic Visualizations**

4. Drag your Sales Trend sheet and drop it right along the bottom edge of the bar chart workspace.



5. Drag your State Profit Map and drop it alongside the right side of the layout



NAME: ADARSH SHUKLA

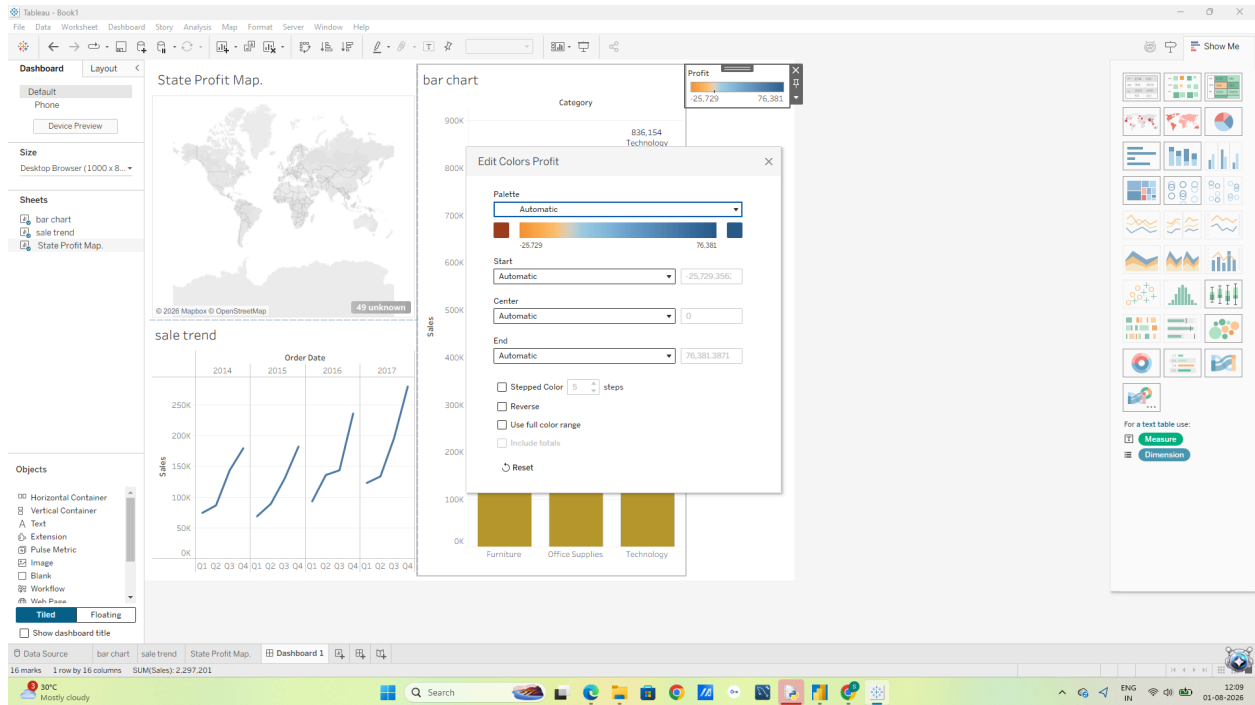
ROLL NO : T117

SHETH L.U.J. AND SIR M.V. COLLEGE OF ARTS, SCIENCE AND COMMERCE

Subject: :Subject: Data Visualization with Power BI and Tableau

Practical 1 :1: **Getting Started with Tableau and Building Basic Visualizations**

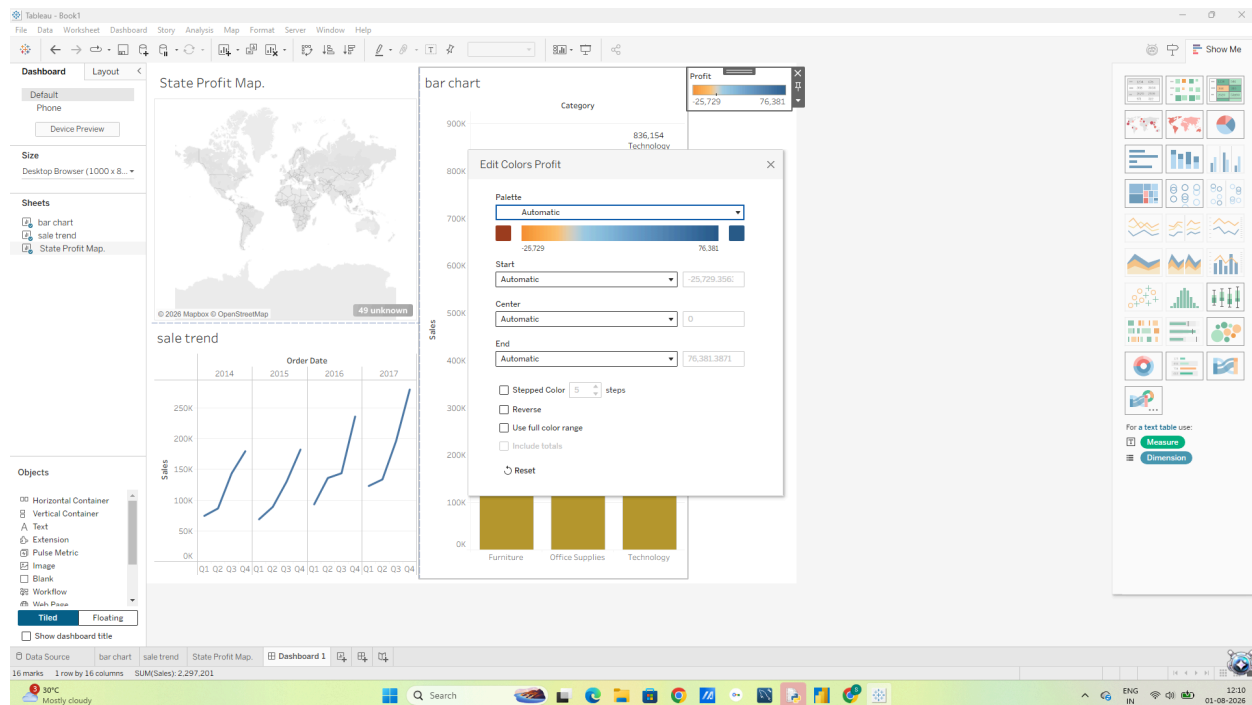
Turning on the Magic: Interactivity 1. Click once inside your Category Bar Chart zone on the dashboard to select



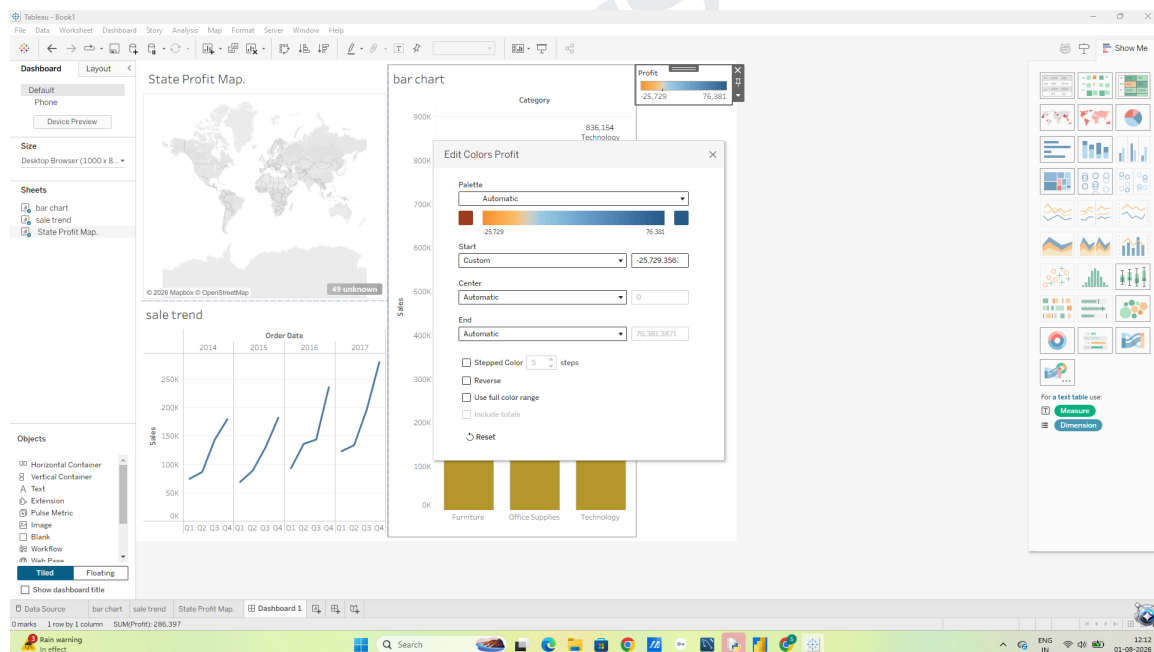
2. Look at the subtle gray menu options border that opens up on its top-right corner marked in red circle.

NAME: ADARSH SHUKLA
ROLL NO : T117

Practical 1 :1: Getting Started with Tableau and Building Basic Visualizations



3. Click the tiny Funnel Icon (Use as Filter).



Now test out your creation! Click on the Technology bar in your bar chart. The map and the time-series line chart will dynamically isolate and update to display data exclusively for your Technology sales. Click the bar a second time to reset the filter

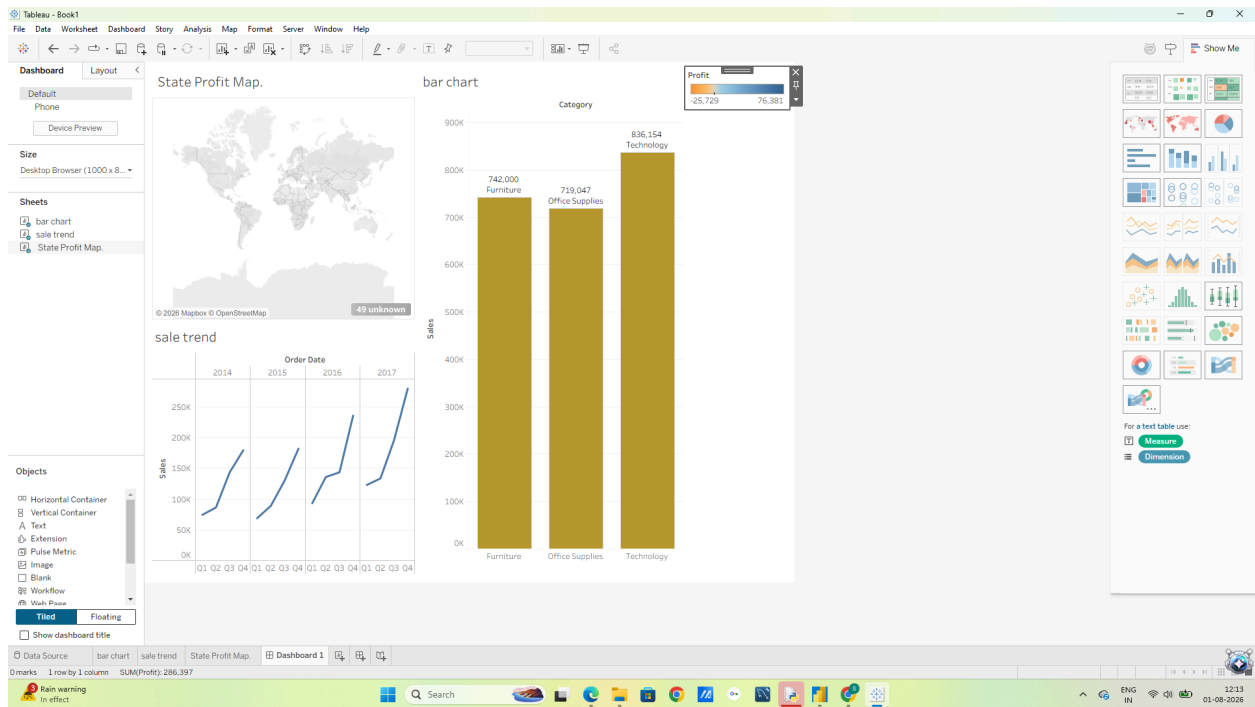
NAME: ADARSH SHUKLA

ROLL NO : T117

SHETH L.U.J. AND SIR M.V. COLLEGE OF ARTS, SCIENCE AND COMMERCE

Subject: :Subject: Data Visualization with Power BI and Tableau

Practical 1 :1: Getting Started with Tableau and Building Basic Visualizations



NAME: ADARSH SHUKLA
ROLL NO : T117